Homework 1

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MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: January 30, 2026

Contents

Problem 1.2

For breakfast Bob has three options: cereal, eggs or fruit. He has to choose exactly two items out of the three available.

(a) Describe the sample space of this experiment.
Hint. What are the different possible outcomes for Bob's breakfast?

(b) Let A be the event that Bob's breakfast includes cereal. Express A as a subset of the sample space.

Solution. We model the experiment by specifying a probability space

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Scratch / Setup. Bob must choose exactly two items from the set

$$\{\text{cereal},\ \text{eggs},\ \text{fruit}\}.$$

Thus, each outcome corresponds to an unordered pair of distinct items.

The sample space is

$$\Omega = \{\{\text{cereal},\ \text{eggs}\},\ \{\text{cereal},\ \text{fruit}\},\ \{\text{eggs},\ \text{fruit}\}\}.$$

We take the σ-field to be the power set $\mathcal{F} = 2^{\Omega}$. Since there is no reason to favor one breakfast over another, we assign equal probability to each outcome.

Each outcome in Ω is equally likely, so

$$\mathbb{P}(\{\omega\}) = \frac{1}{3} \quad \text{for all } \omega \in \Omega.$$

Let A be the event that Bob's breakfast includes cereal. Then

$$A = \{\{\text{cereal},\ \text{eggs}\},\ \{\text{cereal},\ \text{fruit}\}\} \subseteq \Omega.$$

Final: $\boxed{A = \{\{\text{cereal},\ \text{eggs}\},\ \{\text{cereal},\ \text{fruit}\}\}}$

Problem 1.4

Every day a kindergarten class chooses randomly one of the 50 state flags to hang on the wall, without regard to previous choices. We are interested in the flags that are chosen on Monday, Tuesday and Wednesday of next week.

(a) Describe a sample space Ω and a probability measure P to model this experiment.

(b) What is the probability that the class hangs Wisconsin's flag on Monday, Michigan's flag on Tuesday, and California's flag on Wednesday?

(c) What is the probability that Wisconsin's flag will be hung at least two of the three days?

Solution. We model the experiment using a probability space

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Scratch / Setup. Each day (Monday, Tuesday, Wednesday) the class selects one of the 50 state flags, and the choices are made *without regard to previous choices.* Thus we model the outcome as an *ordered* triple of flags.

Let S be the set of the 50 state flags. Define

$$\Omega = S^3 = \{(x_1, x_2, x_3) : x_i \in S\},$$

where x_1 is the flag chosen on Monday, x_2 on Tuesday, and x_3 on Wednesday.

Since Ω is finite, we take the event space to be the power set

$$\mathcal{F} = 2^\Omega.$$

Because each day the choice is uniform over the 50 flags and independent of the other days, all 50^3 outcomes in Ω are equally likely, so

$$\mathbb{P}(\{\omega\}) = \frac{1}{50^3} \quad \text{for all } \omega \in \Omega.$$

Equivalently, for any event $A \subseteq \Omega$,

$$\mathbb{P}(A) = \frac{|A|}{50^3}.$$

(a) The probability space is

$$\Omega = S^3, \qquad \mathcal{F} = 2^\Omega, \qquad \mathbb{P}(\{\omega\}) = \frac{1}{50^3} \text{ for all } \omega \in \Omega.$$

(b) The event that Monday is Wisconsin, Tuesday is Michigan, and Wednesday is California is the singleton

$$\{(\mathrm{WI}, \mathrm{MI}, \mathrm{CA})\} \subseteq \Omega,$$

so
$$\mathbb{P}\big((\mathrm{WI}, \mathrm{MI}, \mathrm{CA})\big) = \frac{1}{50^3}.$$

(c) Let W denote Wisconsin's flag. The event that W appears on at least two of the three days is
$$A = \{(x_1, x_2, x_3) \in S^3 : \#\{i \in \{1,2,3\} : x_i = W\} \geq 2\}.$$

We compute by cases:

- Exactly two days are W: choose which two days ($\binom{3}{2} = 3$ ways), and on the remaining day choose any of the other 49 flags. Thus there are $3 \cdot 49$ outcomes.
- All three days are W: there is 1 outcome.

Hence $|A| = 3 \cdot 49 + 1 = 148$, and therefore
$$\mathbb{P}(A) = \frac{148}{50^3}.$$

Final: $\boxed{\mathbb{P}(A) = \frac{148}{50^3}}$

Problem 1.6

We have an urn with 3 green and 4 yellow balls. We choose 2 balls randomly without replacement. Let A be the event that we have two different colored balls in our sample.

(a) Describe a possible sample space with equally likely outcomes, and the event A in your sample space.

(b) Compute $P(A)$.

Solution. We model the experiment using a probability space

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Label the green balls G_1, G_2, G_3 and the yellow balls Y_1, Y_2, Y_3, Y_4. Since we draw two balls *without replacement* and in sequence, a natural sample space with equally likely outcomes is the set of ordered pairs of distinct balls:

$$\Omega = \{(b_1, b_2) : b_1, b_2 \in \{G_1, G_2, G_3, Y_1, Y_2, Y_3, Y_4\},\ b_1 \neq b_2\}.$$

Then $|\Omega| = 7 \cdot 6 = 42$, we take $\mathcal{F} = 2^\Omega$, and assign

$$\mathbb{P}(\{\omega\}) = \frac{1}{42} \quad \text{for all } \omega \in \Omega.$$

Let A be the event that the two drawn balls have different colors. In this sample space,

$$A = \{(G_i, Y_j) : 1 \leq i \leq 3,\ 1 \leq j \leq 4\} \ \cup\ \{(Y_j, G_i) : 1 \leq j \leq 4,\ 1 \leq i \leq 3\}.$$

Thus

$$|A| = (3 \cdot 4) + (4 \cdot 3) = 24,$$

and therefore

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{24}{42} = \frac{4}{7}.$$

Final: $\boxed{\mathbb{P}(A) = \frac{4}{7}}$

Problem 1.10

We roll a fair die repeatedly until we see the number four appear and then we stop. The outcome of the experiment is the number of rolls.

(a) Following Example 1.16 describe a sample space Ω and a probability measure P to model this situation.

(b) Calculate the probability that the number four never appears.

Solution. We model the experiment using a probability space

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Scratch / Setup. The experiment consists of rolling a fair die repeatedly until the first appearance of the number 4. The outcome is the total number of rolls required.

Thus each outcome is a positive integer $k \in \mathbb{N}$, corresponding to the event that the first $k-1$ rolls are not equal to 4 and the kth roll is equal to 4.

(a) A natural sample space is

$$\Omega = \mathbb{N} = \{1, 2, 3, \dots\}.$$

We take the σ-field $\mathcal{F} = 2^{\Omega}$. Since the die is fair and the rolls are independent, the probability that the first $k-1$ rolls are not 4 and the kth roll is 4 is

$$\mathbb{P}(\{k\}) = \left(\frac{5}{6}\right)^{k-1} \frac{1}{6}, \qquad k \in \Omega.$$

This defines a valid probability measure on Ω.

(b) The event that the number four never appears corresponds to the event that the experiment never terminates. Its probability is

$$\mathbb{P}(\text{no 4 ever appears}) = \lim_{n\to\infty} \mathbb{P}(\text{no 4 in the first } n \text{ rolls}).$$

Since each roll is independent,

$$\mathbb{P}(\text{no 4 in the first } n \text{ rolls}) = \left(\frac{5}{6}\right)^{n}.$$

Taking the limit gives

$$\lim_{n\to\infty} \left(\frac{5}{6}\right)^{n} = 0.$$

Final: $\boxed{\mathbb{P}(\text{four never appears}) = 0}$

Problem 1.11

We throw a dart at a square shaped board of side length 20 inches. Assume that the dart hits the board at a uniformly chosen random point. Find the probability that the dart is within 2 inches of the center of the board.

Solution. We model the experiment by choosing a point uniformly at random from the square board. Since the board has side length 20, its area is

$$\text{Area(board)} = 20 \cdot 20 = 400.$$

The event that the dart is within 2 inches of the center is the disk of radius 2 centered at the board's center. Since $2 < 10$, this disk lies entirely inside the square, and its area is

$$\text{Area(disk)} = \pi \cdot 2^2 = 4\pi.$$

By uniformity, the desired probability is the ratio of areas:

$$\mathbb{P}(\text{within 2 inches of center}) = \frac{4\pi}{400} = \frac{\pi}{100}.$$

Final: $\boxed{\mathbb{P}(\text{within 2 inches of center}) = \frac{\pi}{100}}$

Problem 1.22

We pick a card uniformly at random from a standard deck of 52 cards. (If you are unfamiliar with the deck of 52 cards, see the description above Example C.19 in Appendix C.)

(a) Describe the sample space Ω and the probability measure P that model this experiment.

(b) Give an example of an event in this probability space with probability $\frac{3}{52}$.

(c) Show that there is no event in this probability space with probability $\frac{1}{5}$.

Solution. We model the experiment using a probability space

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

(a) Let D denote the standard deck of 52 cards. A natural sample space is

$$\Omega = D,$$

whose outcomes are the individual cards. Since Ω is finite, we take

$$\mathcal{F} = 2^{\Omega}.$$

Because the card is chosen uniformly at random, each outcome has probability 1/52:

$$\mathbb{P}(\{\omega\}) = \frac{1}{52} \quad \text{for all } \omega \in \Omega.$$

Equivalently, for any event $A \subseteq \Omega$,

Final: $\boxed{\mathbb{P}(A) = \frac{|A|}{52}}$

(b) For example, let

$$A = \{\text{Ace of Spades, Ace of Hearts, Ace of Diamonds}\} \subseteq \Omega.$$

Then $|A| = 3$, so

$$\mathbb{P}(A) = \frac{3}{52}.$$

Final: $\boxed{\text{One example is } A = \{\text{AS}, \text{AH}, \text{AD}\} \text{ with } \mathbb{P}(A) = \dfrac{3}{52}}$

(c) Suppose, for contradiction, that there is an event $B \subseteq \Omega$ with $\mathbb{P}(B) = \frac{1}{5}$. Since every event satisfies $\mathbb{P}(B) = \frac{|B|}{52}$ for some integer $|B| \in \{0, 1, \ldots, 52\}$, we would have

$$\frac{|B|}{52} = \frac{1}{5} \quad \Longrightarrow \quad |B| = \frac{52}{5},$$

which is not an integer. This is a contradiction. Hence no such event exists.

Final: $\boxed{\text{There is no event with probability } \dfrac{1}{5} \text{ in this space.}}$

Problem 1.28

We have an urn with m green balls and n yellow balls. Two balls are drawn at random. What is the probability that the two balls have the same color?

(a) Assume that the balls are sampled without replacement.

(b) Assume that the balls are sampled with replacement.

(c) When is the answer to part (b) larger than the answer to part (a)? Justify your answer. Can you give an intuitive explanation for what the calculation tells you?

Solution. We compute the probability that the two drawn balls have the same color.

(a) (Without replacement.) There are $m+n$ balls total. Using equally likely *unordered* samples of size 2,

$$|\Omega| = \binom{m+n}{2}.$$

The event that the two balls have the same color consists of drawing two green balls or two yellow balls, so the number of favorable outcomes is

$$\binom{m}{2} + \binom{n}{2}.$$

Therefore

$$\mathbb{P}(\text{same color}) = \frac{\binom{m}{2} + \binom{n}{2}}{\binom{m+n}{2}}.$$

Final: $\boxed{\mathbb{P}_{\text{w/o repl.}}(\text{same color}) = \frac{\binom{m}{2} + \binom{n}{2}}{\binom{m+n}{2}}}$

(b) (With replacement.) Now the two draws are independent, and the probability of green on any draw is $\frac{m}{m+n}$ while the probability of yellow is $\frac{n}{m+n}$. Thus

$$\mathbb{P}(\text{same color}) = \left(\frac{m}{m+n}\right)^2 + \left(\frac{n}{m+n}\right)^2 = \frac{m^2+n^2}{(m+n)^2}.$$

Final: $\boxed{\mathbb{P}_{\text{with repl.}}(\text{same color}) = \frac{m^2+n^2}{(m+n)^2}}$

(c) We compare the answers in (a) and (b). Let $T = m+n$ and $S = m^2+n^2$. From (a) and (b),

$$\mathbb{P}_{\text{w/o repl.}} = \frac{\binom{m}{2} + \binom{n}{2}}{\binom{T}{2}} = \frac{S-T}{T(T-1)}, \qquad \mathbb{P}_{\text{with repl.}} = \frac{S}{T^2}.$$

Then

$$\mathbb{P}_{\text{with repl.}} > \mathbb{P}_{\text{w/o repl.}} \iff \frac{S}{T^2} > \frac{S-T}{T(T-1)} \iff S < T^2.$$

But $T^2 = (m+n)^2 = m^2 + 2mn + n^2$, so $S < T^2$ holds exactly when $2mn > 0$, i.e. when $mn > 0$.

Final: $\boxed{\text{The answer in (b) is larger than in (a) exactly when } m > 0 \text{ and } n > 0.}$

Scratch / Setup. *Intuition.* If both colors are present, sampling *without* replacement creates negative dependence: after drawing (say) a green ball, there are fewer green balls left, so the second draw is slightly *less* likely to match the first color. With replacement the composition resets, so matching the first color is more likely; hence the probability of "same color" is larger with replacement.

Homework 2

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MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: February 6, 2026

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Problem 1.12

We roll a fair die repeatedly until we see the number four appear and then we stop.

(a) What is the probability that we need at most 3 rolls?

(b) What is the probability that we needed an even number of die rolls?

Solution.

(a) Let N be the roll on which the first 4 appears. Then N is geometric with $p = 1/6$.

$$P(N \leq 3) = 1 - P(\text{no 4 in first 3 rolls}) = 1 - \left(\frac{5}{6}\right)^3.$$

Final: $\boxed{1 - \left(\frac{5}{6}\right)^3}$

(b)

$$P(N \text{ is even}) = \sum_{k=1}^{\infty} P(N = 2k) = \sum_{k=1}^{\infty} \left(\frac{5}{6}\right)^{2k-1} \frac{1}{6}.$$

This is a geometric series with ratio $(5/6)^2$, so

$$P(N \text{ is even}) = \frac{\frac{1}{6} \cdot \frac{5}{6}}{1 - \left(\frac{5}{6}\right)^2} = \frac{5}{11}.$$

Final: $\boxed{\frac{5}{11}}$

Problem 1.14

Assume that $P(A) = 0.4$ and $P(B) = 0.7$. Making no further assumptions on A and B, show that $P(AB)$ satisfies $0.1 \le P(AB) \le 0.4$.

Solution. By inclusion–exclusion,

$$P(A \cup B) = P(A) + P(B) - P(AB) \le 1,$$

so $P(AB) \ge P(A) + P(B) - 1 = 0.4 + 0.7 - 1 = 0.1$. Also,

$$P(AB) \le \min\{P(A), P(B)\} = \min\{0.4, 0.7\} = 0.4.$$

Final: $\boxed{0.1 \le P(AB) \le 0.4}$

Problem 1.16

We flip a fair coin five times. For every heads you pay me \$1 and for every tails I pay you \$1. Let X denote my net winnings at the end of five flips. Find the possible values and the probability mass function of X.

Solution. Let H be the number of heads in 5 flips. Then $H \sim \text{Binomial}(5, 1/2)$ and

$$X = (\#\text{heads}) - (\#\text{tails}) = H - (5 - H) = 2H - 5.$$

Thus the possible values are

$$X \in \{-5, -3, -1, 1, 3, 5\},$$

with probability mass function

$$P(X = 2h - 5) = P(H = h) = \binom{5}{h} 2^{-5}, \qquad h = 0, 1, 2, 3, 4, 5.$$

Final: $\boxed{P(X = 2h - 5) = \binom{5}{h}/2^5,\ h = 0, \ldots, 5}$

Problem 1.18

The statement

SOME DOGS ARE BROWN

has 16 letters. Choose one of the 16 letters uniformly at random. Let X denote the length of the *word* containing the chosen letter. Determine the possible values and probability mass function of X.

Solution. The words and their lengths are: SOME (4), DOGS (4), ARE (3), BROWN (5). Since 16 letters are equally likely,

$$P(X=3)=\frac{3}{16}, \quad P(X=4)=\frac{8}{16}=\frac{1}{2}, \quad P(X=5)=\frac{5}{16}.$$

Final: $\boxed{X \in \{3,4,5\},\ P(X=3)=\frac{3}{16},\ P(X=4)=\frac{1}{2},\ P(X=5)=\frac{5}{16}}$

Problem 1.32

You are dealt five cards from a standard deck of 52. Find the probability of being dealt a *full house*. (This means that you have three cards of one face value and two cards of a different face value. An example would be three queens and two fours. See Exercise C.10 in Appendix C for a description of all the poker hands.) **Solution.** Count full houses and divide by total hands. Choose the rank of the triple, then its suits, then the rank and suits of the pair:

$$N_{\text{fh}} = 13\binom{4}{3} \cdot 12\binom{4}{2} = 13 \cdot 4 \cdot 12 \cdot 6 = 3744.$$

Total 5-card hands are $\binom{52}{5}$, so

$$P(\text{full house}) = \frac{3744}{\binom{52}{5}} = \frac{6}{4165}.$$

Final: $\boxed{\frac{6}{4165}}$

Problem 1.34

Pick a uniformly chosen random point inside a unit square (a square of sidelength 1) and draw a circle of radius 1/3 around the point. Find the probability that the circle lies entirely inside the square. **Solution.** For the circle of radius 1/3 to lie inside the unit square, the center must be at least 1/3 from each side. Thus the center must lie in

$$\left[\frac{1}{3}, \frac{2}{3}\right] \times \left[\frac{1}{3}, \frac{2}{3}\right],$$

a square of side length $1 - 2 \cdot \frac{1}{3} = \frac{1}{3}$, area 1/9. By uniformity,

$$P(\text{circle inside}) = \frac{1}{9}.$$

Final: $\boxed{\frac{1}{9}}$

Problem 1.36

(a) Let (X, Y) denote a uniformly chosen random point inside the unit square

$$[0,1]^2 = [0,1] \times [0,1] = \{(x,y) : 0 \leq x, y \leq 1\}.$$

Let $0 \leq a < b \leq 1$. Find the probability $P(a < X < b)$, that is, the probability that the x-coordinate X of the chosen point lies in the interval (a, b).

(b) What is the probability $P(|X - Y| \leq 1/4)$?

Solution.

(a) The event $\{a < X < b\}$ corresponds to the vertical strip $(a, b) \times [0, 1]$, which has area $b - a$. Since the point is uniform,

$$P(a < X < b) = b - a.$$

Final: $\boxed{b - a}$

(b) The condition $|X - Y| \leq \frac{1}{4}$ is the band between the lines $y = x \pm \frac{1}{4}$ in the unit square. The excluded regions are two right triangles, each with legs of length $3/4$, so each has area $\frac{1}{2}\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \frac{9}{32}$. Hence

$$P(|X - Y| \leq \tfrac{1}{4}) = 1 - 2 \cdot \frac{9}{32} = 1 - \frac{9}{16} = \frac{7}{16}.$$

Final: $\boxed{\frac{7}{16}}$

Problem 1.40

An urn contains 1 green ball, 1 red ball, 1 yellow ball and 1 white ball. I draw 4 balls with replacement. What is the probability that there is at least one color that is repeated exactly twice?

Hint. Use inclusion–exclusion with events $G = \{\text{exactly two balls are green}\}$, $R = \{\text{exactly two balls are re}$ etc. **Solution.** There are $4^4 = 256$ equally likely sequences. Let G be the event that exactly two balls are green, and similarly R, Y, W. For G, choose the two green positions and fill the other two positions with any of the other 3 colors:

$$|G| = \binom{4}{2} 3^2 = 6 \cdot 9 = 54.$$

Thus $\sum |G| = 4 \cdot 54 = 216$. For a pair, say $G \cap R$, we need exactly two greens and two reds, so

$$|G \cap R| = \binom{4}{2} = 6,$$

and there are $\binom{4}{2} = 6$ such pairs, giving $\sum |G \cap R| = 36$. No triple intersections are possible. By inclusion–exclusion,

$$| \cup G| = 216 - 36 = 180.$$

Therefore

$$P(\text{at least one color appears exactly twice}) = \frac{180}{256} = \frac{45}{64}.$$

Final: $\boxed{\frac{45}{64}}$

Homework 3

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MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: February 13, 2026

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Notation: We sometimes use the standard shorthand for intersections of events: AB means $A \cap B$, A^cB means $A^c \cap B$, and ABC means $A \cap B \cap C$. This shorthand is used consistently throughout.

Problem 2.4

We have two urns. The first urn contains two balls labeled 1 and 2. The second urn contains three balls labeled 3, 4 and 5. We choose one of the urns at random (with equal probability) and then sample one ball (uniformly at random) from the chosen urn. What is the probability that we picked the ball labeled 5?

Solution.

Scratch / Setup. Let U_1 be the event we choose urn 1, U_2 the event we choose urn 2. Let F be the event that we pick ball 5. Use the law of total probability over U_1, U_2.

We have $P(U_1) = P(U_2) = \frac{1}{2}$. Also $P(F \mid U_1) = 0$ and $P(F \mid U_2) = \frac{1}{3}$. Therefore,

$$P(F) = P(F \mid U_1)\, P(U_1) + P(F \mid U_2)\, P(U_2) = 0 \cdot \tfrac{1}{2} + \tfrac{1}{3} \cdot \tfrac{1}{2} = \tfrac{1}{6}.$$

Final: $\boxed{\frac{1}{6}}$

Problem 2.10

I have a bag with 3 fair dice. One is 4-sided, one is 6-sided, and one is 12-sided. I reach into the bag, pick one die at random and roll it. The outcome of the roll is 4. What is the probability that I pulled out the 6-sided die?

Solution.

Scratch / Setup. Let D_4, D_6, D_{12} be the events that the 4-, 6-, or 12-sided die is chosen. Let R be the event that the roll is 4. Use Bayes' rule.

Each die is chosen with probability 1/3. Likelihoods: $P(R \mid D_4) = \frac{1}{4}$, $P(R \mid D_6) = \frac{1}{6}$, $P(R \mid D_{12}) = \frac{1}{12}$. Hence

$$P(D_6 \mid R) = \frac{P(R \mid D_6)\, P(D_6)}{P(R \mid D_4)\, P(D_4) + P(R \mid D_6)\, P(D_6) + P(R \mid D_{12})\, P(D_{12})} = \frac{\frac{1}{6} \cdot \frac{1}{3}}{\frac{1}{4} \cdot \frac{1}{3} + \frac{1}{6} \cdot \frac{1}{3} + \frac{1}{12} \cdot \frac{1}{3}} = \frac{1}{3}.$$

Final: $\boxed{\frac{1}{3}}$

Problem 2.14

Let A and B be two disjoint events. Under what condition are they independent?

Solution.

Scratch / Setup. Disjoint means $A \cap B = \varnothing$, so $P(AB) = 0$. Independence requires $P(AB) = P(A)P(B)$.

Since A and B are disjoint, $P(AB) = 0$. Independence requires $P(A)P(B) = 0$, which happens iff at least one of $P(A)$ or $P(B)$ is zero.

Final: $\boxed{P(A) = 0 \text{ or } P(B) = 0}$

Problem 2.16

We flip a fair coin three times. For $i = 1, 2, 3$, let A_i be the event that among the first i coin flips we have an odd number of heads. Check whether the events A_1, A_2, A_3 are independent or not.

Solution.

Scratch / Setup. Let $\Omega = \{H, T\}^3$ with $|\Omega| = 8$. Compute $P(A_i)$ and all intersections of the A_i.

We have $P(A_1) = P(\text{first flip is } H) = \frac{1}{2}$, $P(A_2) = \frac{1}{2}$ (odd heads among first two), and $P(A_3) = \frac{1}{2}$ (odd heads among three).

Intersections:

$$P(A_1A_2) = P(HT) = \tfrac{1}{4}, \quad P(A_1A_3) = P(\{HHH, HTT\}) = \tfrac{1}{4},$$

$$P(A_2A_3) = P(\{HTT, THT\}) = \tfrac{1}{4}, \quad P(A_1A_2A_3) = P(HTT) = \tfrac{1}{8}.$$

Each pair satisfies $P(A_iA_j) = P(A_i)P(A_j) = \frac{1}{4}$, and the triple satisfies $P(A_1A_2A_3) = P(A_1)P(A_2)P(A_3) = \frac{1}{8}$. Therefore the events are mutually independent.

Final: $\boxed{A_1, A_2, A_3 \text{ are mutually independent}}$

Problem 2.18

We choose a number from the set $\{10, 11, 12, \ldots, 99\}$ uniformly at random.

(a) Let X be the first digit and Y the second digit of the chosen number. Show that X and Y are independent random variables.

(b) Let X be the first digit of the chosen number and Z the sum of the two digits. Show that X and Z are not independent.

Solution.

Scratch / Setup. There are 90 equally likely numbers. For (a), compute $P(X = x)$, $P(Y = y)$, and $P(X = x, Y = y)$. For (b), give a specific (x, z) where factorization fails.

(a) For $x \in \{1, \ldots, 9\}$ and $y \in \{0, \ldots, 9\}$, there is exactly one number with first digit x and second digit y. Thus

$$P(X = x, Y = y) = \tfrac{1}{90}.$$

Also $P(X = x) = \frac{10}{90} = \frac{1}{9}$ and $P(Y = y) = \frac{9}{90} = \frac{1}{10}$. Hence

$$P(X = x, Y = y) = \tfrac{1}{90} = \tfrac{1}{9} \cdot \tfrac{1}{10} = P(X = x)P(Y = y),$$

so X and Y are independent.

(b) Take $x = 1$ and $z = 1$. Then $Z = 1$ occurs only for the number 10, so $P(Z = 1) = \frac{1}{90}$. We have

$$P(X = 1, Z = 1) = \tfrac{1}{90}, \quad P(X = 1)P(Z = 1) = \tfrac{1}{9} \cdot \tfrac{1}{90} = \tfrac{1}{810},$$

which are not equal. Thus X and Z are not independent.

Final: $\boxed{X \text{ and } Y \text{ are independent; } X \text{ and } Z \text{ are not independent}}$

Problem 2.30

Assume that 1/3 of all twins are identical twins. You learn that Miranda is expecting twins, but you have no other information.

(a) Find the probability that Miranda will have two girls.

(b) You learn that Miranda gave birth to two girls. What is the probability that the girls are identical twins?

Explain any assumptions you make.

Solution.

Scratch / Setup. Assume identical twins have the same sex, with boy/girl equally likely; fraternal twins have independent sexes with boy/girl equally likely. Let I be identical, F fraternal, and G be the event of two girls.

We have $P(I) = \frac{1}{3}$ and $P(F) = \frac{2}{3}$. Under the assumptions, $P(G \mid I) = \frac{1}{2}$ and $P(G \mid F) = \frac{1}{4}$.

(a) By total probability,

$$P(G) = P(G \mid I)\, P(I) + P(G \mid F)\, P(F) = \tfrac{1}{2} \cdot \tfrac{1}{3} + \tfrac{1}{4} \cdot \tfrac{2}{3} = \tfrac{1}{3}.$$

Final: $\boxed{P(G) = \frac{1}{3}}$

(b) By Bayes' rule,

$$P(I \mid G) = \frac{P(G \mid I)\, P(I)}{P(G)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{3}} = \tfrac{1}{2}.$$

Final: $\boxed{P(I \mid G) = \frac{1}{2}}$

Problem 2.34

You play the following game against your friend. You have two urns and three balls. One of the balls is marked. You get to place the balls in the two urns any way you please, including leaving one urn empty. Your friend will choose one urn at random and then draw a ball from that urn. (If he chose an empty urn, there is no ball.) His goal is to draw the marked ball.

(a) How would you arrange the balls in the urns to *minimize* his chances of drawing the marked ball?

(b) How would your friend arrange the balls in the urns to *maximize* his chances of drawing the marked ball?

(c) Repeat (a) and (b) for the case of n balls with one marked ball.

Solution.

> **Scratch / Setup.** Let M be the event the marked ball is drawn. If the marked ball is in an urn with k unmarked balls (so $k+1$ total), and the other urn has the remaining balls (possibly empty), then $P(M) = \frac{1}{2} \cdot \frac{1}{k+1}$.

If the marked ball is in an urn with k unmarked balls, then the friend chooses that urn with probability $\frac{1}{2}$ and, conditional on that, draws the marked ball with probability $\frac{1}{k+1}$. Hence

$$P(M) = \frac{1}{2} \cdot \frac{1}{k+1}.$$

(a) To minimize $P(M)$ we maximize k. With three balls total, the best is to put all three balls in one urn and leave the other empty ($k = 2$), giving $P(M) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$.

(b) To maximize $P(M)$ we minimize k, so put the marked ball alone in one urn and the two unmarked balls in the other ($k = 0$), giving $P(M) = \frac{1}{2}$.

(c) With n balls total, the same calculation gives $P(M) = \frac{1}{2} \cdot \frac{1}{k+1}$ when the marked ball is grouped with k unmarked balls. Thus

$$\text{minimize: } k = n - 1 \Rightarrow P(M) = \frac{1}{2n}, \qquad \text{maximize: } k = 0 \Rightarrow P(M) = \frac{1}{2}.$$

Final: $\boxed{\text{minimize } P(M) = \frac{1}{2n} \text{ (all balls together); maximize } P(M) = \frac{1}{2} \text{ (marked ball alone)}}$

Problem 2.52

Suppose that $P(A) = 0.3$, $P(B) = 0.2$, and $P(C) = 0.1$. Further, $P(A \cup B) = 0.44$, $P(A^c C) = 0.07$, $P(BC) = 0.02$, and $P(A \cup B \cup C) = 0.496$. Decide whether A, B, and C are mutually independent.

Solution.

Scratch / Setup. Compute $P(AB)$ from $P(A \cup B)$, compute $P(AC)$ from $P(A^c C)$, and compare pairwise products. Then use inclusion–exclusion to compute $P(ABC)$ and compare to $P(A)P(B)P(C)$.

First,

$$P(AB) = P(A) + P(B) - P(A \cup B) = 0.3 + 0.2 - 0.44 = 0.06.$$

Also $P(A^c C) = 0.07$, so $P(AC) = P(C) - P(A^c C) = 0.1 - 0.07 = 0.03$. We are given $P(BC) = 0.02$.

These match pairwise products: $0.3 \cdot 0.2 = 0.06$, $0.3 \cdot 0.1 = 0.03$, and $0.2 \cdot 0.1 = 0.02$.

Now use inclusion–exclusion:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC).$$

Thus

$$P(ABC) = 0.496 - (0.3 + 0.2 + 0.1) + (0.06 + 0.03 + 0.02) = 0.006.$$

Since $P(A)P(B)P(C) = 0.3 \cdot 0.2 \cdot 0.1 = 0.006$, the triple also factorizes. Hence the events are mutually independent.

Final: $\boxed{A, B, C \text{ are mutually independent}}$

Problem 2.54

Let A and B be events with these properties: $0 < P(B) < 1$ and $P(A|B) = P(A|B^c) = 1/3$.

(a) Is it possible to calculate $P(A)$ from this information? Either declare that it is not possible, or find the value of $P(A)$.

(b) Are A and B independent, not independent, or is it impossible to determine?

Solution.

Scratch / Setup. Use the law of total probability for $P(A)$, then compare $P(A \mid B)$ to $P(A)$ for independence.

(a) By total probability,

$$P(A) = P(A \mid B)\, P(B) + P(A \mid B^c)\, P(B^c) = \tfrac{1}{3}P(B) + \tfrac{1}{3}P(B^c) = \tfrac{1}{3}.$$

Final: $\boxed{P(A) = \tfrac{1}{3}}$

(b) Since $P(A \mid B) = \frac{1}{3} = P(A)$, the events are independent.

Final: $\boxed{A \text{ and } B \text{ are independent}}$

Homework 1

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: February 20, 2026

Contents

Problem 2.20

A fair die is rolled repeatedly. Use precise notation of probabilities of events and random variables for the solutions to the questions below.

(a) Write down a precise sum expression for the probability that the first five rolls give a three at most two times.

(b) Calculate the probability that the first three does not appear before the fifth roll.

(c) Calculate the probability that the first three appears before the twentieth roll but not before the fifth roll.

Solution.

Scratch / Setup. Let S_5 be the number of threes among the first five rolls. Let N be the roll on which the first three appears. Then $S_5 \sim \mathrm{Bin}(5, 1/6)$ and $N \sim \mathrm{Geom}(1/6)$ with support $\{1, 2, \ldots\}$.

(a)

$$P(S_5 \leq 2) = \sum_{k=0}^{2} \binom{5}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{5-k}.$$

(b)

$$P(N > 4) = P(\text{no three in the first four rolls}) = \left(\frac{5}{6}\right)^4.$$

(c)

$$P(5 \leq N \leq 20) = P(\text{no three in the first four rolls}) \cdot P(\text{at least one three in rolls 5–20})$$

$$= \left(\frac{5}{6}\right)^4 \left(1 - \left(\frac{5}{6}\right)^{16}\right) = \left(\frac{5}{6}\right)^4 - \left(\frac{5}{6}\right)^{20}.$$

Final: $\boxed{\text{(a) } \sum_{k=0}^{2} \binom{5}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{5-k}, \text{ (b) } \left(\frac{5}{6}\right)^4, \text{ (c) } \left(\frac{5}{6}\right)^4 - \left(\frac{5}{6}\right)^{20}}$

Problem 2.58

Suppose that a person's birthday is a uniformly random choice from the 365 days of a year (leap years are ignored), and one person's birthday is independent of the birthdays of other people. Alex, Betty and Conlin are comparing birthdays. Define these three events:

$$A = \{\text{Alex and Betty have the same birthday}\},$$

$$B = \{\text{Betty and Conlin have the same birthday}\},$$

$$C = \{\text{Conlin and Alex have the same birthday}\}.$$

(a) Are events A, B and C pairwise independent? (See the definition of pairwise independence on page 54 and Example 2.25 for illustration.)

(b) Are events A, B and C independent?

Solution.

Scratch / Setup. Let birthdays be independent and uniform on $\{1, \ldots, 365\}$. Then $P(A) = P(B) = P(C) = 1/365$. Also $A \cap B$ means all three share a birthday.

(a)

$$P(A \cap B) = P(\text{all three match}) = \frac{1}{365^2} = P(A)P(B).$$

Similarly $P(A \cap C) = P(B \cap C) = 1/365^2$, so A, B, C are pairwise independent.

(b)

$$P(A \cap B \cap C) = \frac{1}{365^2} \neq \left(\frac{1}{365}\right)^3 = P(A)P(B)P(C),$$

so they are not mutually independent.

Final: $\boxed{\text{Pairwise independent, but not mutually independent}}$

Problem 2.61

Suppose an urn has 3 green balls and 4 red balls.

(a) Nine draws are made with replacement. Let X be the number of times a green ball appeared. Identify by name the probability distribution of X. Find the probabilities $P(X \geq 1)$ and $P(X \leq 5)$.

(b) Draws with replacement are made until the first green ball appears. Let N be the number of draws that were needed. Identify by name the probability distribution of N. Find the probability $P(N \leq 9)$.

(c) Compare $P(X \geq 1)$ and $P(N \leq 9)$. Is there a reason these should be the same?

Solution.

Scratch / Setup. Let $p = 3/7$ (green) and $q = 4/7$ (red). For nine draws with replacement, $X \sim \text{Bin}(9, p)$. For the waiting time until the first green, $N \sim \text{Geom}(p)$.

(a) $X \sim \text{Bin}(9, 3/7)$.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{4}{7}\right)^9,$$

$$P(X \leq 5) = \sum_{k=0}^{5} \binom{9}{k} \left(\frac{3}{7}\right)^k \left(\frac{4}{7}\right)^{9-k}.$$

(b) $N \sim \text{Geom}(3/7)$ with support $\{1, 2, \ldots\}$.

$$P(N \leq 9) = 1 - P(N > 9) = 1 - \left(\frac{4}{7}\right)^9.$$

(c) The events $\{X \geq 1\}$ and $\{N \leq 9\}$ both mean "at least one green in the first 9 draws," hence they are equal.

Final: $\boxed{\text{(a) } X \sim \text{Bin}(9, 3/7),\ P(X \geq 1) = 1 - (4/7)^9,\ P(X \leq 5) = \sum_{k=0}^{5} \binom{9}{k} (3/7)^k (4/7)^{9-k}}$

Final: $\boxed{\text{(b) } N \sim \text{Geom}(3/7),\ P(N \leq 9) = 1 - (4/7)^9}$

Problem 2.70

Flip a coin three times. Assume the probability of tails is p and that successive flips are independent. Let A be the event that we have exactly one tails among the first two coin flips and B the event that we have exactly one tails among the last two coin flips. For which values of p are events A and B independent? (This generalizes Example 2.18.)

Solution.

Scratch / Setup. Let $p = P(T)$ and $q = 1 - p$. Then

$$P(A) = P(\text{exactly one tail in flips 1–2}) = 2pq, \qquad P(B) = 2pq.$$

Compute $P(A \cap B)$ by enumerating 3-flip sequences.

The only sequences satisfying both events are HTH and THT, so

$$P(A \cap B) = pq^2 + p^2q = pq.$$

Independence requires $pq = (2pq)^2 = 4p^2q^2$. Thus either $pq = 0$ (i.e., $p = 0$ or $p = 1$) or, for $0 < p < 1$,

$$1 = 4pq \implies p(1-p) = \frac{1}{4} \implies p = \frac{1}{2}.$$

Final: $\boxed{p \in \{0, \tfrac{1}{2}, 1\} \text{ (nontrivial case } p = \tfrac{1}{2})}$

Problem 2.71 (a)-(c)

As in Example 2.38, assume that 90% of the coins in circulation are fair, and the remaining 10% are biased coins that give tails with probability 3/5. I hold a randomly chosen coin and begin to flip it.

(a) After one flip that results in tails, what is the probability that the coin I hold is a biased coin? After two flips that both give tails? After n flips that all come out tails?

(b) After how many straight tails can we say that with 90% probability the coin I hold is biased?

(c) After n straight tails, what is the probability that the next flip is also tails?

(d) Suppose we have flipped a very large number of times (think number of flips n tending to infinity), and each time gotten tails. What are the chances that the next flip again yields tails?

Solution.

Scratch / Setup. Let B be the event the coin is biased and F the event it is fair. $P(B) = 0.1$, $P(F) = 0.9$, $P(T \mid B) = 3/5$, $P(T \mid F) = 1/2$. For n straight tails, use Bayes' rule.

(a) For $n \geq 1$,

$$P(B \mid T^n) = \frac{0.1(3/5)^n}{0.1(3/5)^n + 0.9(1/2)^n} = \frac{1}{1 + 9(5/6)^n}.$$

In particular,

$$P(B \mid T) = \frac{2}{17}, \qquad P(B \mid T^2) = \frac{4}{29}.$$

(b) We need $P(B \mid T^n) \geq 0.9$, i.e.

$$\frac{1}{1 + 9(5/6)^n} \geq 0.9 \iff \left(\frac{5}{6}\right)^n \leq \frac{1}{81}.$$

The smallest integer satisfying this is $n = 25$.

(c) The predictive probability is

$$P(T_{n+1} \mid T^n) = \frac{1}{2}\big(1 - P(B \mid T^n)\big) + \frac{3}{5}P(B \mid T^n) = \frac{1}{2} + \frac{1}{10} \cdot \frac{1}{1 + 9(5/6)^n}.$$

Part (d) not assigned for HW4.

Final: $\boxed{\text{(a) } P(B \mid T^n) = \frac{1}{1+9(5/6)^n}}$

Final: $\boxed{\text{(b) } n = 25}$

Final: $\boxed{\text{(c) } P(T_{n+1} \mid T^n) = \frac{1}{2} + \frac{1}{10} \cdot \frac{1}{1+9(5/6)^n}}$

Problem 3.2

Suppose the random variable X has possible values $\{1, 2, 3, 4, 5, 6\}$ and probability mass function of the form $p(k) = ck$.

(a) Find c.

(b) Find the probability that X is odd.

Solution.

Scratch / Setup. Normalize: $\sum_{k=1}^{6} ck = 1$.

(a)

$$c = \frac{1}{1+2+3+4+5+6} = \frac{1}{21}.$$

(b)

$$P(X \text{ odd}) = c(1+3+5) = \frac{9}{21} = \frac{3}{7}.$$

Final: $\boxed{c = \frac{1}{21}, \quad P(X \text{ odd}) = \frac{3}{7}}$

Problem 3.4

Let $X \sim \text{Unif}[4, 10]$.

(a) Calculate $P(X < 6)$.

(b) Calculate $P(|X - 7| > 1)$.

(c) For $4 \leq t \leq 6$, calculate $P(X < t \mid X < 6)$.

Solution.

Scratch / Setup. For $X \sim \text{Unif}[4, 10]$, $P(X \in [a, b]) = (b - a)/6$.

(a)

$$P(X < 6) = \frac{6 - 4}{10 - 4} = \frac{1}{3}.$$

(b) $|X - 7| > 1$ means $X < 6$ or $X > 8$:

$$P(|X - 7| > 1) = \frac{2 + 2}{6} = \frac{2}{3}.$$

(c) For $4 \leq t \leq 6$,

$$P(X < t \mid X < 6) = \frac{P(4 \leq X < t)}{P(4 \leq X < 6)} = \frac{t - 4}{2}.$$

Final: $\boxed{\text{(a) } \frac{1}{3}, \text{ (b) } \frac{2}{3}, \text{ (c) } \frac{t-4}{2}}$

Problem 3.25

In each of the following cases find all values of b for which the given function is a probability density function.

(a)

$$f(x) = \begin{cases} x^2 - b, & \text{if } 1 \le x \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

(b)

$$h(x) = \begin{cases} \cos x, & \text{if } -b \le x \le b, \\ 0, & \text{otherwise.} \end{cases}$$

Solution.

Scratch / Setup. For a p.d.f., we need nonnegativity on the support and total integral $= 1$.

(a)

$$\int_1^3 (x^2 - b)\,dx = \frac{26}{3} - 2b = 1 \implies b = \frac{23}{6}.$$

But nonnegativity requires $x^2 - b \ge 0$ for $x \in [1,3]$, hence $b \le 1$. These conditions are incompatible, so no value of b works.

(b)

$$\int_{-b}^{b} \cos x\,dx = 2\sin b = 1 \implies \sin b = \frac{1}{2}.$$

To keep $\cos x \ge 0$ on $[-b, b]$ we need $b \le \pi/2$, so $b = \pi/6$.

Final: $\boxed{\text{(a) no solution,} \quad \text{(b) } b = \frac{\pi}{6}}$

Problem 3.26 (a)

Suppose that X is a discrete random variable with possible values $\{1, 2, \dots\}$, and probability mass function

$$p_X(k) = \frac{c}{k(k+1)}$$

with some constant $c > 0$.

(a) What is the value of c?

Hint. $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$.

Solution.

Scratch / Setup. Use the telescoping sum $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$.

$$1 = \sum_{k=1}^{\infty} \frac{c}{k(k+1)} = c \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right) = c.$$

Final: $\boxed{c = 1}$

Problem 3.31 (a)-(d)

Suppose a random variable X has density function

$$f(x) = \begin{cases} cx^{-4}, & \text{if } x \geq 1, \\ 0, & \text{else,} \end{cases}$$

where c is a constant.

(a) What must be the value of c?

(b) Find $P(0.5 < X < 1)$.

(c) Find $P(0.5 < X < 2)$.

(d) Find $P(2 < X < 4)$.

Solution.

Scratch / Setup. Normalize f on $[1, \infty)$ and integrate for the probabilities.

(a)

$$1 = \int_1^\infty cx^{-4}\,dx = c \cdot \frac{1}{3} \quad \Longrightarrow \quad c = 3.$$

(b) Since the support is $[1, \infty)$,

$$P(0.5 < X < 1) = 0.$$

(c)

$$P(0.5 < X < 2) = \int_1^2 3x^{-4}\,dx = 3\left[\frac{-1}{3}x^{-3}\right]_1^2 = \frac{7}{8}.$$

(d)

$$P(2 < X < 4) = \int_2^4 3x^{-4}\,dx = 3\left[\frac{-1}{3}x^{-3}\right]_2^4 = \frac{7}{64}.$$

Final: $\boxed{\text{(a) } c = 3, \text{ (b) } 0, \text{ (c) } \frac{7}{8}, \text{ (d) } \frac{7}{64}}$

Homework 5

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: February 27, 2026

Contents

Problem 3.5

Suppose that the discrete random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 0, & \text{if } x < 1, \\ \frac{1}{3}, & \text{if } 1 \leq x < \frac{4}{3}, \\ \frac{1}{2}, & \text{if } \frac{4}{3} \leq x < \frac{3}{2}, \\ \frac{3}{4}, & \text{if } \frac{3}{2} \leq x < \frac{9}{5}, \\ 1, & \text{if } x \geq \frac{9}{5}. \end{cases}$$

Find the possible values and the probability mass function X.

Solution.

Scratch / Setup. The c.d.f. jumps at the support points. Each point mass equals the jump size: $P(X = x) = F(x) - \lim_{t \uparrow x} F(t)$.

The jump points are $1, \frac{4}{3}, \frac{3}{2}, \frac{9}{5}$, so these are the possible values. The masses are

$$P(X = 1) = \frac{1}{3} - 0 = \frac{1}{3}, \qquad P\left(X = \frac{4}{3}\right) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6},$$

$$P\left(X = \frac{3}{2}\right) = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}, \qquad P\left(X = \frac{9}{5}\right) = 1 - \frac{3}{4} = \frac{1}{4}.$$

These sum to 1.

Final: $\boxed{X \in \left\{1, \frac{4}{3}, \frac{3}{2}, \frac{9}{5}\right\},\ P(X = 1) = \frac{1}{3},\ P\big(X = \frac{4}{3}\big) = \frac{1}{6},\ P\big(X = \frac{3}{2}\big) = \frac{1}{4},\ P\big(X = \frac{9}{5}\big) = \frac{1}{4}}$

Problem 3.7

Suppose that the continuous random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 0, & \text{if } x < \sqrt{2}, \\ x^2 - 2, & \text{if } \sqrt{2} \le x < \sqrt{3}, \\ 1, & \text{if } \sqrt{3} \le x. \end{cases}$$

(a) Find the smallest interval $[a, b]$ such that of $P(a \le X \le b) = 1$.

(b) Find $P(X = 1.6)$.

(c) Find $P(1 \le X \le \frac{3}{2})$.

(d) Find the probability density function of X.

Solution.

Scratch / Setup. Support is where F rises from 0 to 1, i.e., $[\sqrt{2}, \sqrt{3}]$. Since X is continuous, point probabilities are 0. The pdf is $f = F'$ on the interval where F is differentiable.

(a) The smallest interval with probability 1 is

$$[a, b] = \left[\sqrt{2}, \sqrt{3}\right].$$

(b) Since X is continuous,

$$P(X = 1.6) = 0.$$

(c)

$$P\left(1 \le X \le \frac{3}{2}\right) = F\left(\frac{3}{2}\right) - F(1)$$

and $1 < \sqrt{2}$, so $F(1) = 0$ and

$$F\left(\frac{3}{2}\right) = \left(\frac{3}{2}\right)^2 - 2 = \frac{1}{4}.$$

Thus $P(1 \le X \le 3/2) = 1/4$.

(d)

$$f_X(x) = F'(x) = \begin{cases} 2x, & \sqrt{2} < x < \sqrt{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Final: $\boxed{\text{(a) } [\sqrt{2}, \sqrt{3}],\ \text{(b) } 0,\ \text{(c) } \frac{1}{4},\ \text{(d) } f_X(x) = \begin{cases} 2x, & \sqrt{2} < x < \sqrt{3} \\ 0, & \text{otherwise} \end{cases}}$

Problem 3.9

Let X be the random variable from Exercise 3.3.

(a) Find the mean of X.

(b) Compute $E[e^{2X}]$.

Solution.

Scratch / Setup. From Exercise 3.3, $f(x) = 3e^{-3x}$ for $x > 0$ (and 0 otherwise), so $X \sim \text{Exp}(3)$. Use $E[X] = \int_0^\infty x f(x)\,dx$ and $E[e^{2X}] = \int_0^\infty e^{2x} f(x)\,dx$.

(a)

$$E[X] = \int_0^\infty x \cdot 3e^{-3x}\,dx = \frac{1}{3}.$$

(b)

$$E[e^{2X}] = \int_0^\infty e^{2x} \cdot 3e^{-3x}\,dx = 3\int_0^\infty e^{-x}\,dx = 3.$$

Final: $\boxed{\text{(a) } E[X] = \tfrac{1}{3}, \quad \text{(b) } E[e^{2X}] = 3}$

Problem 3.10

Let X have probability mass function

$$P(X = -1) = \frac{1}{2}, \quad P(X = 0) = \frac{1}{3}, \quad \text{and } P(X = 1) = \frac{1}{6}.$$

Calculate $E[|X|]$ using the approaches in (a) and (b) below.

(a) First find the probability mass function of the random variable $Y = |X|$ and using that compute $E[|X|]$.

(b) Apply formula (3.24) with $g(x) = |x|$.

Solution.

Scratch / Setup. Let $Y = |X|$. Then $Y \in \{0, 1\}$ with $P(Y = 0) = P(X = 0) = 1/3$ and $P(Y = 1) = P(X = 1) + P(X = -1) = 2/3$.

(a) The p.m.f. of Y is

$$P(Y = 0) = \frac{1}{3}, \qquad P(Y = 1) = \frac{2}{3}.$$

Thus

$$E[|X|] = E[Y] = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}.$$

(b) Using (3.24) with $g(x) = |x|$,

$$E[|X|] = \sum_x |x| P(X = x) = 1 \cdot \frac{1}{2} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{6} = \frac{2}{3}.$$

Final: $\boxed{E[|X|] = \frac{2}{3}}$

Problem 3.47

Choose a point uniformly at random from the triangle with vertices $(0,0)$, $(30,0)$, and $(30,20)$. Let (X,Y) be the coordinates of the chosen point.

(a) Find the cumulative distribution function of X.

(b) Use part (a) to find the density function of X.

(c) Use part (b) to find the expectation of X.

Solution.

Scratch / Setup. The triangle has area $\frac{1}{2}\cdot 30\cdot 20 = 300$. The upper boundary line from $(0,0)$ to $(30,20)$ is $y=\frac{2}{3}x$. For $0\le x\le 30$, the area with $X\le x$ is $\int_0^x \frac{2}{3}t\,dt = \frac{1}{3}x^2$.

(a)

$$F_X(x) = \begin{cases} 0, & x<0, \\ \dfrac{x^2}{900}, & 0\le x<30, \\ 1, & x\ge 30. \end{cases}$$

(b) Differentiate:

$$f_X(x) = \begin{cases} \dfrac{x}{450}, & 0\le x<30, \\ 0, & \text{otherwise.} \end{cases}$$

(c)

$$E[X] = \int_0^{30} x\, f_X(x)\,dx = \int_0^{30} x\cdot\frac{x}{450}\,dx = \frac{1}{450}\cdot\frac{30^3}{3} = 20.$$

Final: $\boxed{\text{(a) } F_X(x) = \begin{cases} 0, & x<0 \\ x^2/900, & 0\le x<30 \\ 1, & x\ge 30 \end{cases}}$

Final: $\boxed{\text{(b) } f_X(x) = \begin{cases} x/450, & 0\le x<30 \\ 0, & \text{otherwise} \end{cases}}$

Final: $\boxed{\text{(c) } E[X]=20}$

Problem 3.48

Pick a random point uniformly inside the triangle with vertices $(0,0)$, $(2,0)$ and $(0,1)$. Compute the expectation of the distance of this point to the y-axis.

Solution.

Scratch / Setup. The distance to the y-axis equals the x-coordinate. The triangle has area $\frac{1}{2} \cdot 2 \cdot 1 = 1$ and upper boundary $y = 1 - \frac{x}{2}$ for $0 \leq x \leq 2$.

Let R be the distance to the y-axis; then $R = X$. Thus

$$E[R] = \frac{1}{\text{Area}} \int_0^2 x \left(1 - \frac{x}{2}\right) dx = \int_0^2 \left(x - \frac{x^2}{2}\right) dx = \left[\frac{x^2}{2} - \frac{x^3}{6}\right]_0^2 = \frac{2}{3}.$$

Final: $\boxed{\frac{2}{3}}$

Homework 6

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: March 6, 2026

Contents

Problem 3.16

Let Z have the following density function

$$f_Z(z) = \begin{cases} \frac{1}{7}, & 1 \le z \le 2, \\ \frac{3}{7}, & 5 \le z \le 7, \\ 0, & \text{otherwise.} \end{cases}$$

Compute both $E[Z]$ and $\operatorname{Var}(Z)$.

Solution.

Scratch / Setup. Support: $z \in [1,2] \cup [5,7]$, with $f_Z(z) = \frac{1}{7}$ on $[1,2]$ and $f_Z(z) = \frac{3}{7}$ on $[5,7]$. Plan: compute $E[Z] = \int z f_Z(z)\,dz$ and $E[Z^2] = \int z^2 f_Z(z)\,dz$ over those intervals, then $\operatorname{Var}(Z) = E[Z^2] - (E[Z])^2$.

$$E[Z] = \int_{-\infty}^{\infty} z f_Z(z)\,dz = \frac{1}{7}\int_1^2 z\,dz + \frac{3}{7}\int_5^7 z\,dz = \frac{1}{7}\cdot\frac{2^2-1^2}{2} + \frac{3}{7}\cdot\frac{7^2-5^2}{2} = \frac{75}{14}.$$

$$E[Z^2] = \int_{-\infty}^{\infty} z^2 f_Z(z)\,dz = \frac{1}{7}\int_1^2 z^2\,dz + \frac{3}{7}\int_5^7 z^2\,dz = \frac{1}{7}\cdot\frac{2^3-1^3}{3} + \frac{3}{7}\cdot\frac{7^3-5^3}{3} = \frac{661}{21}.$$

$$\operatorname{Var}(Z) = E[Z^2] - (E[Z])^2 = \frac{661}{21} - \left(\frac{75}{14}\right)^2 = \frac{1633}{588}.$$

Final: $\boxed{E[Z] = \frac{75}{14},\ \operatorname{Var}(Z) = \frac{1633}{588}}$

Problem 3.18

Let X be a normal random variable with mean 3 and variance 4.

(a) Find the probability $P(2 < X < 6)$.

(b) Find the value c such that $P(X > c) = 0.33$.

(c) Find $E(X^2)$.

Hint. You can integrate with the density function, but it is quicker to relate $E(X^2)$ to the mean and variance.

Solution.

Scratch / Setup. $\mu = 3$, $\sigma = 2$. Standardize: $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ with CDF Φ. (a) Compute $\Phi(1.5) - \Phi(-0.5)$. (b) Solve $\Phi(\frac{c-3}{2}) = 0.67$, so $c = 3 + 2z_{0.67}$. (c) Use $E(X^2) = \mathrm{Var}(X) + (E[X])^2$.

(a)

$$P(2 < X < 6) = P\left(\frac{2-3}{2} < Z < \frac{6-3}{2}\right) = P\left(-\tfrac{1}{2} < Z < \tfrac{3}{2}\right) = \Phi(1.5) - \Phi(-0.5).$$

Using $\Phi(1.5) \approx 0.9332$ and $\Phi(0.5) \approx 0.6915$,

$$P(2 < X < 6) \approx 0.9332 - (1 - 0.6915) = 0.6247.$$

(b) We need c such that

$$0.33 = P(X > c) = P\left(Z > \frac{c-3}{2}\right) = 1 - \Phi\left(\frac{c-3}{2}\right).$$

Thus $\Phi(\frac{c-3}{2}) = 0.67$. Let $z_{0.67}$ satisfy $\Phi(z_{0.67}) = 0.67$; from the table $z_{0.67} \approx 0.44$. Hence $c = 3 + 2z_{0.67} \approx 3 + 2(0.44) = 3.88$.

(c)

$$E(X^2) = \mathrm{Var}(X) + (E[X])^2 = 4 + 3^2 = 13.$$

Final: $\boxed{P(2 < X < 6) \approx 0.6247,\ c \approx 3.88,\ E(X^2) = 13}$

Problem 3.23

Ten thousand people each buy a lottery ticket. Each lottery ticket costs \$1. 100 people are chosen as winners. Of those 100 people, 80 will win \$2, 19 will win \$100, and one lucky winner will win \$7000. Let X denote the profit (profit = winnings – cost) of a randomly chosen player of this game.

(a) Give both the possible values and probability mass function for X.

(b) Find $P(X \geq 100)$.

(c) Compute $E[X]$ and $\mathrm{Var}(X)$.

Solution.

Scratch / Setup. Possible winnings: $0, 2, 100, 7000$. Profit X = winnings – 1 so values are $-1, 1, 99, 6999$. Probabilities: 9900 nonwinners, 80 win \$2, 19 win \$100, 1 win \$7000 out of 10,000. Compute $E[X]$, $E[X^2]$, then $\mathrm{Var}(X) = E[X^2] - (E[X])^2$.

(a) There are 10,000 tickets and 100 winners. Hence

$$P(X=-1)=\frac{9900}{10000}=\frac{99}{100}, \quad P(X=1)=\frac{80}{10000}=\frac{1}{125},$$
$$P(X=99)=\frac{19}{10000}, \quad P(X=6999)=\frac{1}{10000}.$$

(b) Since $99 < 100$, the only value with $X \geq 100$ is 6999:

$$P(X \geq 100) = P(X = 6999) = \frac{1}{10000}.$$

(c)

$$E[X]=(-1)\frac{99}{100}+1\cdot\frac{1}{125}+99\cdot\frac{19}{10000}+6999\cdot\frac{1}{10000}=-\frac{47}{500}\approx -0.094.$$
$$E[X^2]=1\cdot\frac{99}{100}+1\cdot\frac{1}{125}+99^2\cdot\frac{19}{10000}+6999^2\cdot\frac{1}{10000}=\frac{245911}{50}=4918.22.$$
$$\mathrm{Var}(X)=E[X^2]-(E[X])^2=\frac{245911}{50}-\left(\frac{47}{500}\right)^2\approx 4918.211.$$

Final: $\boxed{E[X]=-\frac{47}{500},\ \mathrm{Var}(X)\approx 4918.211,\ P(X\geq 100)=\frac{1}{10000}}$

Problem 3.28

There are five closed boxes on a table. Three of the boxes have nice prizes inside. The other two do not. You open boxes one at a time until you find a prize. Let X be the number of boxes you open.

(a) Find the probability mass function of X.

(b) Find $E[X]$.

(c) Find $\mathrm{Var}(X)$.

(d) Suppose the prize inside each of the three good boxes is \$100, but each empty box you open costs you \$100. What is your expected gain or loss in this game?

Hint. Express the gain or loss as a function of X.

Solution.

Scratch / Setup. X = number of boxes opened until first prize. Possible values: $1, 2, 3$. $\mathrm{P}(X = 1)$: first box good. $\mathrm{P}(X = 2)$: first empty, then good. $\mathrm{P}(X = 3)$: two empties, then good. Then compute $E[X]$, $E[X^2]$, and $\mathrm{Var}(X) = E[X^2] - (E[X])^2$. For (d) let gain/loss $W = 100(2 - X)$ and use $E[aX + b] = aE[X] + b$.

(a)

$$P(X = 1) = \frac{3}{5}, \quad P(X = 2) = \frac{2}{5} \cdot \frac{3}{4} = \frac{3}{10}, \quad P(X = 3) = \frac{2}{5} \cdot \frac{1}{4} \cdot \frac{3}{3} = \frac{1}{10}.$$

(b)

$$E[X] = 1 \cdot \frac{3}{5} + 2 \cdot \frac{3}{10} + 3 \cdot \frac{1}{10} = \frac{3}{2}.$$

(c)

$$E[X^2] = 1^2 \cdot \frac{3}{5} + 2^2 \cdot \frac{3}{10} + 3^2 \cdot \frac{1}{10} = \frac{27}{10},$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{27}{10} - \left(\frac{3}{2}\right)^2 = \frac{9}{20}.$$

(d) Each empty box costs \$100 and the prize is \$100, so the gain/loss is

$$W = 100 - 100(X - 1) = 100(2 - X).$$

By linearity,

$$E[W] = 100(2 - E[X]) = 100\left(2 - \frac{3}{2}\right) = 50.$$

Final: $\boxed{P(X = 1, 2, 3) = \left(\frac{3}{5}, \frac{3}{10}, \frac{1}{10}\right),\ E[X] = \frac{3}{2},\ \mathrm{Var}(X) = \frac{9}{20},\ E[W] = 50}$

Problem 3.66

Let X be a normal random variable with mean 8 and variance 3. Find the value of α such that $P(X > \alpha) = 0.15$.

Solution.

Scratch / Setup. $\mu = 8$, $\sigma = \sqrt{3}$. Let $Z = \frac{X-8}{\sqrt{3}} \sim N(0,1)$. Solve $P(Z > (\alpha - 8)/\sqrt{3}) = 0.15$, so $\Phi((\alpha - 8)/\sqrt{3}) = 0.85$.

$$0.15 = P(X > \alpha) = P\left(Z > \frac{\alpha - 8}{\sqrt{3}}\right) = 1 - \Phi\left(\frac{\alpha - 8}{\sqrt{3}}\right).$$

Thus $\Phi\left(\frac{\alpha-8}{\sqrt{3}}\right) = 0.85$. From the table, $z_{0.85} \approx 1.04$. Therefore

$$\alpha = 8 + \sqrt{3}\, z_{0.85} \approx 8 + \sqrt{3}(1.04) \approx 9.80.$$

Final: $\boxed{\alpha = 8 + \sqrt{3}\, z_{0.85} \approx 9.80}$

Problem 3.71

My bus is scheduled to depart at noon. However, in reality the departure time varies randomly, with average departure time 12 o'clock noon and a standard deviation of 6 minutes. Assume the departure time is normally distributed. If I get to the bus stop 5 minutes past noon, what is the probability that the bus has not yet departed?

Solution.

Scratch / Setup. Let X be the departure time in minutes relative to noon. Then $X \sim N(0, 36)$ and $Z = X/6 \sim N(0,1)$. We need $P(X > 5) = P(Z > 5/6)$ and relate to Φ.

$$P(\text{bus not yet departed}) = P(X > 5) = P\left(Z > \frac{5}{6}\right) = 1 - \Phi\left(\frac{5}{6}\right).$$

Using $\Phi(0.83) \approx 0.7967$,

$$P(X > 5) \approx 1 - 0.7967 = 0.2033.$$

Final: $\boxed{P(X > 5) = 1 - \Phi(5/6) \approx 0.2033}$

Homework 7

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: March 13, 2026

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Problem 4.4

Hint. You will need the continuity correction for this. Liz is standing on the real number line at position 0. She rolls a die repeatedly. If the roll is 1 or 2, she takes *one* step to the right (in the positive direction). If the roll is 3, 4, 5 or 6, she takes *two* steps to the right. Let X_n be Liz's position after n rolls of the die. Estimate the probability that X_{90} is at least 160.

Solution.

Scratch / Setup. Let Y_i be the step on roll i: $Y_i = 1$ with probability 1/3 (roll 1 or 2) and $Y_i = 2$ with probability 2/3 (roll 3–6). Then $X_n = \sum_{i=1}^n Y_i$. Let S_n be the number of two-steps. Then $S_n \sim \mathrm{Bin}(n, 2/3)$ and $X_n = n + S_n$. For $n = 90$, $E[S_{90}] = 60$, $\mathrm{Var}(S_{90}) = 20$, $\sigma = \sqrt{20}$. We need $P(X_{90} \geq 160) = P(S_{90} \geq 70)$ and use normal approximation with continuity correction.

Since $X_{90} = 90 + S_{90}$,

$$P(X_{90} \geq 160) = P(S_{90} \geq 70) \approx P\left(\frac{S_{90} - 60}{\sqrt{20}} \geq \frac{69.5 - 60}{\sqrt{20}}\right).$$

Thus

$$P(X_{90} \geq 160) \approx 1 - \Phi(2.12) \approx 0.0168.$$

Final: $\boxed{P(X_{90} \geq 160) \approx 0.0168}$

Problem 4.6

A pollster would like to estimate the fraction p of people in a population who intend to vote for a particular candidate. How large must a random sample be in order to be at least 95% certain that the fraction $\hat{p}$ of positive answers in the sample is within 0.02 of the true p?

Solution.

Scratch / Setup. Let $S_n \sim \mathrm{Bin}(n,p)$ be the number of positive answers and $\hat{p} = S_n/n$. We want $P(|\hat{p}-p| \le 0.02) \ge 0.95$. Use the normal approximation for $\hat{p}$: $P(|\hat{p}-p| < \varepsilon) \approx 2\Phi\left(\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1$. Since $p(1-p) \le 1/4$, a conservative bound is $P(|\hat{p}-p| < \varepsilon) \gtrsim 2\Phi(2\varepsilon\sqrt{n}) - 1$.

With $\varepsilon = 0.02$, require

$$2\Phi(0.04\sqrt{n}) - 1 \ge 0.95 \quad \Longrightarrow \quad \Phi(0.04\sqrt{n}) \ge 0.975.$$

Using $z_{0.975} = 1.96$,

$$0.04\sqrt{n} \ge 1.96 \quad \Longrightarrow \quad \sqrt{n} \ge 49 \quad \Longrightarrow \quad n \ge 2401.$$

Final: $\boxed{n \ge 2401}$

Problem 4.8

In a million rolls of a biased die the number 6 shows up 180,000 times. Find a 99.9% confidence interval for the unknown probability that the die rolls 6.

Solution.

Scratch / Setup. Let $S \sim \text{Bin}(n, p)$ be the number of 6's in $n = 1{,}000{,}000$ rolls and $\hat{p} = S/n = 0.18$. For a 99.9% confidence interval, we want $P(|\hat{p} - p| < \varepsilon) \approx 0.999$. Normal approximation gives $P(|\hat{p} - p| < \varepsilon) \approx 2\Phi\left(\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1$. Using $\sqrt{p(1-p)} \leq 1/2$ yields the conservative bound $P(|\hat{p} - p| < \varepsilon) \gtrsim 2\Phi(2\varepsilon\sqrt{n}) - 1$. Set $2\Phi(2\varepsilon\sqrt{n}) - 1 = 0.999$ so $2\varepsilon\sqrt{n} = z_{0.9995} \approx 3.29$.

Thus

$$\varepsilon \approx \frac{3.29}{2\sqrt{1{,}000{,}000}} = \frac{3.29}{2000} \approx 0.00165,$$

and the 99.9% confidence interval is

$$\hat{p} \pm \varepsilon = 0.18 \pm 0.00165 = (0.17835,\ 0.18165).$$

Final: $\boxed{(0.17835,\ 0.18165)}$

Problem 4.10

A hockey player scores at least one goal in roughly half of his games. How would you estimate the percentage of games where he scores a hat-trick (three goals)?

Solution.

Scratch / Setup. Model goals per game by $G \sim \text{Poisson}(\lambda)$ (rare scoring chances). Then $P(G \geq 1) = 1 - e^{-\lambda} \approx 0.5$ so $\lambda = \ln 2$. A hat-trick means exactly three goals, so compute $P(G = 3)$.

With $\lambda = \ln 2$,

$$P(G = 3) = e^{-\lambda}\frac{\lambda^3}{3!} = \frac{(\ln 2)^3}{6}e^{-\ln 2} = \frac{(\ln 2)^3}{12} \approx 0.0278.$$

So the estimate is about 2.8% of games.

Final: $\boxed{P(\text{hat-trick}) \approx 0.0278}$

Problem 4.16

We choose 500 numbers uniformly at random from the interval $[1.5, 4.8]$.

(a) Approximate the probability of the event that less than 65 of the numbers start with the digit 1.

(b) Approximate the probability of the event that more than 160 of the numbers start with the digit 3.

Solution.

Scratch / Setup. A number in $[1.5, 4.8]$ starts with digit 1 iff it lies in $[1.5, 2)$; length 0.5. So $p_1 = 0.5/(4.8 - 1.5) = 5/33$. Let $S \sim \text{Bin}(500, p_1)$ be the count. For digit 3, the interval is $[3, 4)$ of length 1, so $p_3 = 1/3.3 = 10/33$. Let $T \sim \text{Bin}(500, p_3)$. Use normal approximation with continuity correction.

(a) $\mu_S = 500 \cdot \frac{5}{33} = \frac{2500}{33} \approx 75.76$ and $\sigma_S^2 = 500 \cdot \frac{5}{33} \cdot \frac{28}{33} \approx 64.28$, so $\sigma_S \approx 8.02$. With continuity correction,

$$P(S < 65) = P(S \leq 64) \approx \Phi\left(\frac{64.5 - 75.76}{8.02}\right) = \Phi(-1.40) \approx 0.080.$$

(b) $\mu_T = 500 \cdot \frac{10}{33} = \frac{5000}{33} \approx 151.52$ and $\sigma_T^2 = 500 \cdot \frac{10}{33} \cdot \frac{23}{33} \approx 105.6$, so $\sigma_T \approx 10.28$. With continuity correction,

$$P(T > 160) = P(T \geq 161) \approx 1 - \Phi\left(\frac{160.5 - 151.52}{10.28}\right) = 1 - \Phi(0.87) \approx 0.191.$$

Final: $\boxed{P(S < 65) \approx 0.080, \quad P(T > 160) \approx 0.191}$

Problem 4.26

100 randomly chosen individuals were interviewed to estimate the unknown fraction $p \in (0, 1)$ of the population that prefers whole milk to skim milk. The resulting estimate is $\hat{p}$. With what level of confidence can we state that the true p lies in the interval $(\hat{p} - 0.1, \hat{p} + 0.1)$?

Solution.

Scratch / Setup. Let $S \sim \text{Bin}(n, p)$ with $n = 100$, and $\hat{p} = S/n$. We want the confidence level $P(|\hat{p} - p| < 0.1)$. Normal approximation gives $P(|\hat{p} - p| < \varepsilon) \approx 2\Phi\left(\frac{\varepsilon\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1$. Since $p(1-p) \le 1/4$, a lower bound is $2\Phi(2\varepsilon\sqrt{n}) - 1$.

With $\varepsilon = 0.1$ and $n = 100$,

$$P(|\hat{p} - p| < 0.1) \approx 2\Phi(0.2\sqrt{100}) - 1 = 2\Phi(2) - 1 \approx 0.9545.$$

So the confidence level is about 95.45%.

Final: $\boxed{\text{confidence} \approx 0.9545}$

Homework 8

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: March 20, 2026

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Problem 4.34

A taxi company has a large fleet of cars. On average, there are 3 accidents each week. What is the probability that at most 2 accidents happen next week? Make some reasonable assumption in order to be able to answer the question. Simplify your answer as much as possible.

Solution.

Scratch / Setup. Let X be the number of accidents next week. A standard modeling assumption is: each taxi has a small probability of an accident during the week, accidents for different taxis are approximately independent, and the fleet is large. Then a Poisson model is appropriate, with parameter equal to the weekly mean:

$$X \sim \text{Poisson}(3).$$

Assuming accidents occur as independent rare events across a large fleet, the number of accidents in one week is well modeled by a Poisson random variable with mean 3. Thus

$$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2).$$

Using the Poisson p.m.f.,

$$P(X \le 2) = e^{-3}\left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!}\right) = e^{-3}\left(1 + 3 + \frac{9}{2}\right) = \frac{17}{2}e^{-3}.$$

Final: $\boxed{P(X \le 2) = \frac{17}{2}e^{-3}}$

Problem 4.42

On average 20% of the gadgets produced by a factory are mildly defective. I buy a box of 100 gadgets. Assume this is a random sample from the production of the factory. Let A be the event that less than 15 gadgets in the random sample of 100 are mildly defective.

(a) Give an exact expression for $P(A)$, without attempting to evaluate it.

(b) Use either the normal or the Poisson approximation, whichever is appropriate, to give an approximation of $P(A)$.

Solution.

Scratch / Setup. Let X be the number of mildly defective gadgets in the box. Then

$$X \sim \text{Bin}(100, 0.2), \qquad A = \{X < 15\} = \{X \leq 14\}.$$

For approximation, compare Poisson and normal:

$$np = 20, \qquad np(1-p) = 16.$$

The normal approximation is the better choice here. Using continuity correction,

$$P(X \leq 14) \approx P\left(Z \leq \frac{14.5 - 20}{4}\right) = P(Z \leq -1.375).$$

(a) Since $X \sim \text{Bin}(100, 0.2)$,

$$P(A) = P(X \leq 14) = \sum_{k=0}^{14} \binom{100}{k} (0.2)^k (0.8)^{100-k}.$$

(b) Because the mean $np = 20$ and variance $np(1-p) = 16$ are both reasonably large, the normal approximation is more appropriate than the Poisson approximation. Approximating X by $N(20, 16)$ and using continuity correction,

$$P(A) = P(X \leq 14) \approx P\left(Z \leq \frac{14.5 - 20}{4}\right) = P(Z \leq -1.375).$$

Numerically,

$$P(A) \approx \Phi(-1.375) \approx 0.0846.$$

Final: $\boxed{\text{(a) } \sum_{k=0}^{14} \binom{100}{k} (0.2)^k (0.8)^{100-k}, \qquad \text{(b) } P(A) \approx \Phi(-1.375) \approx 0.0846}$

Problem 4.46

Jessica flips a fair coin 5 times every morning, for 30 days straight. Let X be the number of mornings over these 30 days on which all 5 flips are tails. Use either the normal or the Poisson approximation, whichever is more appropriate, to give an estimate for the probability $P(X = 2)$. Justify your choice of approximation.

Solution.

Scratch / Setup. For one morning, the probability of getting 5 tails in 5 flips is

$$p = \left(\frac{1}{2}\right)^5 = \frac{1}{32}.$$

If X is the number of such mornings in 30 days, then

$$X \sim \text{Bin}\left(30, \frac{1}{32}\right).$$

Here

$$np = \frac{15}{16}, \qquad np^2 = \frac{15}{512},$$

so the event is rare and the Poisson approximation is more appropriate than the normal approximation.

Let X be the number of mornings, out of 30, on which all 5 flips are tails. Then

$$X \sim \text{Bin}\left(30, \frac{1}{32}\right).$$

Since $p = 1/32$ is small and $np^2 = 15/512$ is very small, the Poisson approximation is appropriate. We approximate X by

$$Y \sim \text{Poisson}\left(\lambda = \frac{30}{32}\right) = \text{Poisson}\left(\frac{15}{16}\right).$$

Therefore,

$$P(X = 2) \approx P(Y = 2) = e^{-15/16}\frac{(15/16)^2}{2!} = \frac{225}{512}e^{-15/16}.$$

Numerically,

$$P(X = 2) \approx 0.1721.$$

Final: $\boxed{P(X = 2) \approx \frac{225}{512}e^{-15/16} \approx 0.1721}$

Problem 5.2

Suppose that X has moment generating function

$$M_X(t) = \frac{1}{2} + \frac{1}{3}e^{-4t} + \frac{1}{6}e^{5t}.$$

(a) Find the mean and variance of X by differentiating the moment generating function to find moments.

(b) Find the probability mass function of X. Use the probability mass function to check your answer for part (a).

Solution.

Scratch / Setup. Differentiate the mgf to get moments:

$$M_X'(t) = -\frac{4}{3}e^{-4t} + \frac{5}{6}e^{5t}, \qquad M_X''(t) = \frac{16}{3}e^{-4t} + \frac{25}{6}e^{5t}.$$

Also, because

$$M_X(t) = \sum_x P(X = x)e^{tx},$$

the exponential terms show the support and probabilities directly.

(a) Using the mgf,

$$E[X] = M_X'(0) = -\frac{4}{3} + \frac{5}{6} = -\frac{8}{6} + \frac{5}{6} = -\frac{1}{2}.$$

Also,

$$E[X^2] = M_X''(0) = \frac{16}{3} + \frac{25}{6} = \frac{32}{6} + \frac{25}{6} = \frac{57}{6} = \frac{19}{2}.$$

Hence

$$\mathrm{Var}(X) = E[X^2] - \big(E[X]\big)^2 = \frac{19}{2} - \left(-\frac{1}{2}\right)^2 = \frac{19}{2} - \frac{1}{4} = \frac{37}{4}.$$

(b) Since

$$M_X(t) = \frac{1}{2} + \frac{1}{3}e^{-4t} + \frac{1}{6}e^{5t},$$

we read off the distribution:

$$P(X = 0) = \frac{1}{2}, \qquad P(X = -4) = \frac{1}{3}, \qquad P(X = 5) = \frac{1}{6}.$$

Checking part (a) from the p.m.f.,

$$E[X] = (-4)\cdot\frac{1}{3} + 0\cdot\frac{1}{2} + 5\cdot\frac{1}{6} = -\frac{4}{3} + \frac{5}{6} = -\frac{1}{2},$$

which agrees with the mgf calculation.

Final: $\boxed{E[X] = -\tfrac{1}{2}, \quad \mathrm{Var}(X) = \tfrac{37}{4}, \quad P(X = -4) = \tfrac{1}{3},\ P(X = 0) = \tfrac{1}{2},\ P(X = 5) = \tfrac{1}{6}}$

Problem 5.8

Let $X \sim \text{Unif}[-1, 2]$. Find the probability density function of the random variable $Y = X^2$.

Solution.

Scratch / Setup. Let $Y = X^2$. Since $X \sim \text{Unif}[-1, 2]$, its density is

$$f_X(x) = \frac{1}{3} \quad \text{for } -1 \le x \le 2.$$

The map $x \mapsto x^2$ is not one-to-one: for $0 < y < 1$, the preimages are $\pm\sqrt{y}$; for $1 < y < 4$, only $\sqrt{y}$ lies in $[-1, 2]$. Use

$$f_Y(y) = \sum_{x:\, x^2 = y} f_X(x) \frac{1}{|2x|}.$$

The possible values of $Y = X^2$ are $0 \le Y \le 4$.

If $0 < y < 1$, both $\sqrt{y}$ and $-\sqrt{y}$ belong to the support of X, so

$$f_Y(y) = f_X(\sqrt{y})\frac{1}{2\sqrt{y}} + f_X(-\sqrt{y})\frac{1}{2\sqrt{y}} = \frac{1}{3} \cdot \frac{1}{2\sqrt{y}} + \frac{1}{3} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{3\sqrt{y}}.$$

If $1 < y < 4$, only $\sqrt{y}$ belongs to $[-1, 2]$, so

$$f_Y(y) = f_X(\sqrt{y})\frac{1}{2\sqrt{y}} = \frac{1}{3} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{6\sqrt{y}}.$$

Outside $(0, 4)$ the density is 0. Therefore,

$$f_Y(y) = \begin{cases} \frac{1}{3\sqrt{y}}, & 0 < y < 1, \\ \frac{1}{6\sqrt{y}}, & 1 < y < 4, \\ 0, & \text{otherwise.} \end{cases}$$

The value at $y = 1$ is irrelevant for the density.

Final: $\boxed{f_Y(y) = \begin{cases} \frac{1}{3\sqrt{y}}, & 0 < y < 1, \\ \frac{1}{6\sqrt{y}}, & 1 < y < 4, \\ 0, & \text{otherwise} \end{cases}}$

Problem 5.10

Suppose that X has moment generating function

$$M(t) = \left(\frac{1}{5} + \frac{4}{5}e^t\right)^{30}.$$

What is the distribution of X?

Hint. Do Exercise 5.9 first.

Solution.

Scratch / Setup. From Exercise 5.9, the mgf of a binomial random variable is

$$M(t) = (1 - p + pe^t)^n.$$

Compare this with

$$\left(\frac{1}{5} + \frac{4}{5}e^t\right)^{30}.$$

By the binomial mgf formula,

$$M_X(t) = (1 - p + pe^t)^n$$

for $X \sim \mathrm{Bin}(n, p)$. Since

$$M(t) = \left(\frac{1}{5} + \frac{4}{5}e^t\right)^{30},$$

we identify

$$n = 30, \qquad p = \frac{4}{5}.$$

Therefore,

$$X \sim \mathrm{Bin}\left(30, \frac{4}{5}\right).$$

Equivalently, its p.m.f. is

$$P(X = k) = \binom{30}{k}\left(\frac{4}{5}\right)^k\left(\frac{1}{5}\right)^{30-k}, \qquad k = 0, 1, \ldots, 30.$$

Final: $\boxed{X \sim \mathrm{Bin}\left(30, \tfrac{4}{5}\right)}$

Problem 5.18

Let $X \sim \text{Geom}(p)$.

(a) Compute the moment generating function $M_X(t)$ of X. Be careful about the possibility that $M_X(t)$ might be infinite.

(b) Use the moment generating function to compute the mean and the variance of X.

Solution.

Scratch / Setup. Write $q = 1 - p$. Then for $X \sim \text{Geom}(p)$ on $\{1, 2, \dots\}$,

$$P(X = k) = pq^{k-1}.$$

The mgf becomes a geometric series:

$$M_X(t) = \sum_{k=1}^{\infty} e^{tk} pq^{k-1} = pe^t \sum_{k=0}^{\infty} (qe^t)^k.$$

(a) For those t such that $qe^t < 1$, the series converges and

$$M_X(t) = pe^t \sum_{k=0}^{\infty} (qe^t)^k = \frac{pe^t}{1 - qe^t} = \frac{pe^t}{1 - (1-p)e^t}.$$

Thus

$$M_X(t) = \begin{cases} \dfrac{pe^t}{1 - (1-p)e^t}, & t < \ln\left(\dfrac{1}{1-p}\right), \\ \infty, & t \geq \ln\left(\dfrac{1}{1-p}\right). \end{cases}$$

(b) Differentiate

$$M_X(t) = \frac{pe^t}{1 - qe^t}.$$

Then

$$M_X'(t) = \frac{pe^t}{(1 - qe^t)^2},$$

so

$$E[X] = M_X'(0) = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}.$$

Differentiating once more,

$$M_X''(t) = \frac{pe^t}{(1 - qe^t)^2} + \frac{2pq\, e^{2t}}{(1 - qe^t)^3},$$

hence

$$E[X^2] = M_X''(0) = \frac{1}{p} + \frac{2(1-p)}{p^2} = \frac{2-p}{p^2}.$$

Therefore,

$$\mathrm{Var}(X) = E[X^2] - \big(E[X]\big)^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}.$$

Final: $\boxed{M_X(t) = \frac{pe^t}{1-(1-p)e^t} \text{ for } t < \ln\!\left(\frac{1}{1-p}\right), \quad E[X] = \frac{1}{p}, \quad \mathrm{Var}(X) = \frac{1-p}{p^2}}$

Problem 5.20

Suppose that random variable X has density function $f(x) = \frac{1}{2}e^{-|x|}$.

(a) Compute the moment generating function $M_X(t)$ of X. Be careful about the possibility that $M_X(t)$ might be infinite.

(b) Use the moment generating function to compute the nth moment of X.

Solution.

Scratch / Setup. Split the integral at 0. For $x \geq 0$, $|x| = x$; for $x < 0$, $|x| = -x$. So

$$M_X(t) = \frac{1}{2}\int_0^\infty e^{tx}e^{-x}\,dx \;+\; \frac{1}{2}\int_{-\infty}^0 e^{tx}e^{x}\,dx.$$

Both integrals converge exactly when $-1 < t < 1$.

(a) We compute

$$M_X(t) = \frac{1}{2}\int_0^\infty e^{-(1-t)x}\,dx + \frac{1}{2}\int_{-\infty}^0 e^{(1+t)x}\,dx.$$

The first integral converges when $t < 1$ and equals $\dfrac{1}{2(1-t)}$. The second converges when $t > -1$ and equals $\dfrac{1}{2(1+t)}$. Hence

$$M_X(t) = \begin{cases} \dfrac{1}{2(1-t)} + \dfrac{1}{2(1+t)} = \dfrac{1}{1-t^2}, & -1 < t < 1, \\ \infty, & |t| \geq 1. \end{cases}$$

(b) For $|t| < 1$,

$$M_X(t) = \frac{1}{1-t^2} = \sum_{k=0}^\infty t^{2k}.$$

Comparing with

$$M_X(t) = \sum_{n=0}^\infty \frac{E[X^n]}{n!}t^n,$$

we see that odd powers do not appear, so

$$E[X^n] = 0 \qquad \text{for odd } n.$$

If $n = 2k$ is even, the coefficient of t^{2k} is 1, so

$$\frac{E[X^{2k}]}{(2k)!} = 1, \qquad \text{hence} \qquad E[X^{2k}] = (2k)!.$$

Final: $\boxed{M_X(t) = \frac{1}{1-t^2} \text{ for } |t| < 1, \qquad E[X^n] = \begin{cases} 0, & n \text{ odd}, \\ n!, & n \text{ even} \end{cases}}$

Problem 5.24

Let $X \sim N(0,1)$ and $Y = e^X$. Y is called a *log-normal* random variable.

(a) Find the probability density function of Y.

(b) Find the nth moment $E(Y^n)$ of Y.

Hint. Do not compute the moment generating function of Y. Instead relate the nth moment of Y to an expectation of X that you know.

Solution.

Scratch / Setup. Since $Y = e^X$ and the exponential function is increasing, use the cdf method:

$$F_Y(y) = P(Y \le y) = P(e^X \le y).$$

Also $Y > 0$ always. For moments,

$$Y^n = e^{nX},$$

so $E[Y^n]$ is the standard normal mgf evaluated at n.

(a) Because $Y = e^X > 0$, we have $F_Y(y) = 0$ for $y \le 0$. For $y > 0$,

$$F_Y(y) = P(e^X \le y) = P(X \le \ln y) = \Phi(\ln y),$$

where Φ is the standard normal cdf. Differentiating,

$$f_Y(y) = \Phi'(\ln y) \cdot \frac{1}{y} = \frac{1}{\sqrt{2\pi}} e^{-(\ln y)^2/2} \cdot \frac{1}{y}, \qquad y > 0.$$

Thus

$$f_Y(y) = \begin{cases} \dfrac{1}{y\sqrt{2\pi}} e^{-(\ln y)^2/2}, & y > 0, \\ 0, & y \le 0. \end{cases}$$

(b) Since $Y = e^X$,

$$E[Y^n] = E[e^{nX}] = M_X(n).$$

For $X \sim N(0,1)$,

$$M_X(t) = e^{t^2/2},$$

so

$$E[Y^n] = e^{n^2/2}.$$

Final: $\boxed{f_Y(y) = \begin{cases} \dfrac{1}{y\sqrt{2\pi}} e^{-(\ln y)^2/2}, & y > 0, \\ 0, & y \le 0 \end{cases}, \qquad E[Y^n] = e^{n^2/2}}$

Problem 5.32

Suppose that X is a uniform random variable on the interval $(0, 1)$ and let $Y = 1/X$. Find the probability density function of Y.

Solution.

Scratch / Setup. Use the cdf. Since $0 < X < 1$, the transformed variable is

$$Y = \frac{1}{X} > 1.$$

For $y > 1$,

$$F_Y(y) = P\left(\frac{1}{X} \le y\right) = P\left(X \ge \frac{1}{y}\right).$$

Because X is uniform on $(0, 1)$, its density is 1 on $(0, 1)$. Since $Y = 1/X$, the support of Y is $(1, \infty)$.

If $y \le 1$, then $F_Y(y) = 0$. If $y > 1$, then

$$F_Y(y) = P\left(\frac{1}{X} \le y\right) = P\left(X \ge \frac{1}{y}\right) = 1 - \frac{1}{y}.$$

Differentiating for $y > 1$ gives

$$f_Y(y) = \frac{d}{dy}\left(1 - \frac{1}{y}\right) = \frac{1}{y^2}.$$

Therefore,

$$f_Y(y) = \begin{cases} \frac{1}{y^2}, & y > 1, \\ 0, & y \le 1. \end{cases}$$

Final: $\boxed{f_Y(y) = \begin{cases} \frac{1}{y^2}, & y > 1, \\ 0, & y \le 1 \end{cases}}$

Homework 9

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

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Contents

Problem 6.2

The joint probability mass function of the random variables (X, Y) is given by the following table:

$X \backslash Y$	0	1	2	3
1	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
2	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{10}$
3	$\frac{1}{30}$	$\frac{1}{30}$	0	$\frac{1}{10}$

(a) Find the marginal probability mass functions of X and Y.

(b) Calculate the probability $P(X + Y^2 \le 2)$.

Solution.

Scratch / Setup. Let $p_{X,Y}(x, y)$ denote the table entries. For part (a),

$$p_X(x) = \sum_y p_{X,Y}(x, y), \qquad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

For part (b), we list the pairs with $x + y^2 \le 2$. Since $y^2 \ge 4$ when $y = 2$ or 3, only $y = 0, 1$ can contribute.

(a) Summing the rows gives

$$p_X(1) = \frac{1}{15} + \frac{1}{15} + \frac{2}{15} + \frac{1}{15} = \frac{1}{3},$$

$$p_X(2) = \frac{1}{10} + \frac{1}{10} + \frac{1}{5} + \frac{1}{10} = \frac{1}{2},$$

$$p_X(3) = \frac{1}{30} + \frac{1}{30} + 0 + \frac{1}{10} = \frac{1}{6}.$$

So the marginal p.m.f. of X is

x	1	2	3
$p_X(x)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

Summing the columns gives

$$p_Y(0) = \frac{1}{15} + \frac{1}{10} + \frac{1}{30} = \frac{1}{5}, \qquad p_Y(1) = \frac{1}{15} + \frac{1}{10} + \frac{1}{30} = \frac{1}{5},$$

$$p_Y(2) = \frac{2}{15} + \frac{1}{5} + 0 = \frac{1}{3}, \qquad p_Y(3) = \frac{1}{15} + \frac{1}{10} + \frac{1}{10} = \frac{4}{15}.$$

Hence

y	0	1	2	3
$p_Y(y)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{3}$	$\frac{4}{15}$

(b) The condition $x + y^2 \leq 2$ is satisfied only for

$$(x, y) = (1, 0),\ (1, 1),\ (2, 0).$$

Therefore,

$$P(X + Y^2 \leq 2) = p_{X,Y}(1, 0) + p_{X,Y}(1, 1) + p_{X,Y}(2, 0) = \frac{1}{15} + \frac{1}{15} + \frac{1}{10} = \frac{7}{30}.$$

Final: $\boxed{\text{(a) } p_X(1) = \frac{1}{3},\ p_X(2) = \frac{1}{2},\ p_X(3) = \frac{1}{6}}$

Final: $\boxed{\text{(a) } p_Y(0) = \frac{1}{5},\ p_Y(1) = \frac{1}{5},\ p_Y(2) = \frac{1}{3},\ p_Y(3) = \frac{4}{15}}$

Final: $\boxed{\text{(b) } P(X + Y^2 \leq 2) = \frac{7}{30}}$

Problem 6.6

Suppose that X, Y are jointly continuous with joint probability density function

$$f(x,y) = \begin{cases} xe^{-x(1+y)}, & \text{if } x > 0 \text{ and } y > 0 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the marginal density functions of X and Y.

(b) Calculate the expectation $E[XY]$.

(c) Calculate the expectation $E\left[\frac{X}{1+Y}\right]$.

Solution.

Scratch / Setup. The support is $x > 0$, $y > 0$. For the marginals,

$$f_X(x) = \int_0^\infty xe^{-x(1+y)}\,dy, \qquad f_Y(y) = \int_0^\infty xe^{-x(1+y)}\,dx.$$

Useful integrals:

$$\int_0^\infty e^{-ay}\,dy = \frac{1}{a}, \quad \int_0^\infty xe^{-ax}\,dx = \frac{1}{a^2}, \quad \int_0^\infty x^2e^{-ax}\,dx = \frac{2}{a^3} \qquad (a > 0).$$

(a) For $x > 0$,

$$f_X(x) = \int_0^\infty xe^{-x(1+y)}\,dy = xe^{-x}\int_0^\infty e^{-xy}\,dy = xe^{-x}\cdot\frac{1}{x} = e^{-x}.$$

Thus

$$f_X(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

For $y > 0$,

$$f_Y(y) = \int_0^\infty xe^{-x(1+y)}\,dx = \frac{1}{(1+y)^2}.$$

Hence

$$f_Y(y) = \begin{cases} \dfrac{1}{(1+y)^2}, & y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

(b)

$$E[XY] = \int_0^\infty\int_0^\infty xy\,xe^{-x(1+y)}\,dy\,dx = \int_0^\infty x^2e^{-x}\left(\int_0^\infty ye^{-xy}\,dy\right)dx.$$

Since $\int_0^\infty ye^{-xy}\,dy = \frac{1}{x^2}$ for $x > 0$,

$$E[XY] = \int_0^\infty e^{-x}\,dx = 1.$$

(c)
$$E\left[\frac{X}{1+Y}\right] = \int_0^\infty \int_0^\infty \frac{x}{1+y}\, x e^{-x(1+y)}\, dy\, dx.$$

Integrating first in x,

$$E\left[\frac{X}{1+Y}\right] = \int_0^\infty \frac{1}{1+y} \left(\int_0^\infty x^2 e^{-x(1+y)}\, dx \right) dy$$

$$= \int_0^\infty \frac{1}{1+y} \cdot \frac{2}{(1+y)^3}\, dy = 2\int_0^\infty \frac{1}{(1+y)^4}\, dy = \frac{2}{3}.$$

Final: $\boxed{\text{(a) } f_X(x) = e^{-x}\mathbf{1}_{\{x>0\}}, \quad f_Y(y) = \frac{1}{(1+y)^2}\mathbf{1}_{\{y>0\}}}$

Final: $\boxed{\text{(b) } E[XY] = 1, \qquad \text{(c) } E\left[\frac{X}{1+Y}\right] = \frac{2}{3}}$

Problem 6.8

Determine whether the following pairs of random variables X and Y are independent.

(a) Random variables X and Y of Exercise 6.2.

(b) Random variables X and Y of Exercise 6.5.

(d) Random variables X and Y of Exercise 6.7.

Solution.

Scratch / Setup. For independence we compare the joint distribution with the product of the marginals.

- In part (a), it is enough to find one pair (x, y) with $p_{X,Y}(x, y) \neq p_X(x)p_Y(y)$.
- In part (b), use the marginals from Exercise 6.5: $f_X(x) = \frac{6}{7}x + \frac{4}{7}$ and $f_Y(y) = \frac{12}{7}y^2 + \frac{6}{7}y$ on $[0, 1]$.
- In part (d), the support is the triangle $\{(x, y) : 0 \leq x,\ 0 \leq y,\ x + y \leq 1\}$, not a rectangle.

(a) From Exercise 6.2,

$$P(X = 3) = \frac{1}{6} > 0, \qquad P(Y = 2) = \frac{1}{3} > 0,$$

but from the table,

$$P(X = 3, Y = 2) = 0.$$

Hence $P(X = 3, Y = 2) \neq P(X = 3)P(Y = 2)$, so X and Y are not independent.

(b) From Exercise 6.5,

$$f_{X,Y}(x, y) = \frac{12}{7}(xy + y^2), \qquad 0 \leq x \leq 1,\ 0 \leq y \leq 1,$$

and the marginals are

$$f_X(x) = \frac{6}{7}x + \frac{4}{7}, \qquad f_Y(y) = \frac{12}{7}y^2 + \frac{6}{7}y.$$

Evaluating at $x = y = \frac{1}{4}$,

$$f_{X,Y}\left(\frac{1}{4}, \frac{1}{4}\right) = \frac{12}{7}\left(\frac{1}{16} + \frac{1}{16}\right) = \frac{3}{14},$$

while

$$f_X\left(\frac{1}{4}\right) f_Y\left(\frac{1}{4}\right) = \left(\frac{11}{14}\right)\left(\frac{9}{28}\right) = \frac{99}{392} \neq \frac{3}{14}.$$

So X and Y are not independent.

(c) In Exercise 6.7, (X, Y) is uniform on the triangle

$$\{(x, y) : 0 \leq x,\ 0 \leq y,\ x + y \leq 1\}.$$

The marginal densities are positive on $(0, 1)$:

$$f_X(x) = 2(1 - x), \qquad f_Y(y) = 2(1 - y).$$

Take $x = y = \frac{3}{4}$. Then

$$f_X\left(\frac{3}{4}\right) > 0, \qquad f_Y\left(\frac{3}{4}\right) > 0,$$

but

$$f_{X,Y}\left(\frac{3}{4}, \frac{3}{4}\right) = 0$$

because $\frac{3}{4} + \frac{3}{4} > 1$, so the point lies outside the triangle. Therefore $f_{X,Y}(x, y) \neq f_X(x) f_Y(y)$, and X and Y are not independent.

Final: $\boxed{\text{(a), (b), and (d) are all not independent}}$

Problem 6.18

Suppose that X and Y are integer-valued random variables with joint probability mass function given by

$$p_{X,Y}(a,b) = \begin{cases} \dfrac{1}{4a}, & \text{for } 1 \le b \le a \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Show that this is indeed a joint probability mass function.

(b) Find the marginal probability mass functions of X and Y.

(c) Find $P(X = Y + 1)$.

Solution.

Scratch / Setup. The support is

$$\{(a,b) \in \mathbb{Z}^2 : 1 \le b \le a \le 4\}.$$

For part (a),

$$\sum_{a=1}^{4}\sum_{b=1}^{a} \frac{1}{4a} = \sum_{a=1}^{4} \frac{a}{4a} = 1.$$

For the marginals,

$$p_X(a) = \sum_{b=1}^{a} p_{X,Y}(a,b), \qquad p_Y(b) = \sum_{a=b}^{4} p_{X,Y}(a,b).$$

(a) Since $p_{X,Y}(a,b) \ge 0$ on its support, it remains only to check that the total mass is 1:

$$\sum_{a=1}^{4}\sum_{b=1}^{a} \frac{1}{4a} = \sum_{a=1}^{4} \frac{a}{4a} = 4 \cdot \frac{1}{4} = 1.$$

So this is a valid joint probability mass function.

(b) For $a = 1, 2, 3, 4$,

$$p_X(a) = \sum_{b=1}^{a} \frac{1}{4a} = a \cdot \frac{1}{4a} = \frac{1}{4}.$$

Thus

$$p_X(a) = \frac{1}{4}, \qquad a = 1, 2, 3, 4.$$

For Y,

$$p_Y(1) = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{25}{48},$$
$$p_Y(2) = \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48},$$

$$p_Y(3) = \frac{1}{12} + \frac{1}{16} = \frac{7}{48}, \qquad p_Y(4) = \frac{1}{16}.$$

(c) The event $\{X = Y + 1\}$ consists of the admissible pairs

$$(2,1),\ (3,2),\ (4,3).$$

Therefore,

$$P(X = Y + 1) = \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{6+4+3}{48} = \frac{13}{48}.$$

Final: $\boxed{\text{(a) valid joint pmf,} \quad p_X(a) = \frac{1}{4} \text{ for } a = 1,2,3,4}$

Final: $\boxed{p_Y(1) = \frac{25}{48},\ p_Y(2) = \frac{13}{48},\ p_Y(3) = \frac{7}{48},\ p_Y(4) = \frac{1}{16}}$

Final: $\boxed{\text{(c) } P(X = Y + 1) = \frac{13}{48}}$

Problem 6.24

An urn contains 1 green ball, 1 red ball, 1 yellow ball and 1 white ball. I draw 3 balls with replacement. What is the probability that exactly two balls are of the same color?

Solution.

Scratch / Setup. Treat each draw as a color outcome from $\{G, R, Y, W\}$. There are

$$4^3 = 64$$

equally likely ordered color sequences. For a favorable outcome, one color appears exactly twice and a different color appears once.

To get exactly two balls of one color and one ball of a different color:

$$\text{choose the repeated color: 4 ways,}$$

$$\text{choose the singleton color: 3 ways,}$$

$$\text{choose the position of the singleton: 3 ways.}$$

Hence the number of favorable ordered outcomes is

$$4 \cdot 3 \cdot 3 = 36.$$

Since there are $4^3 = 64$ possible ordered outcomes in total,

$$P(\text{exactly two balls are of the same color}) = \frac{36}{64} = \frac{9}{16}.$$

Final: $\boxed{\frac{9}{16}}$

Problem 6.27

Suppose that X_1 and X_2 are independent random variables with

$$P(X_1 = 1) = P(X_1 = -1) = \frac{1}{2}$$

and

$$P(X_2 = 1) = 1 - P(X_2 = -1) = p$$

with $0 < p < 1$. Let $Y = X_1 X_2$. Show that X_2 and Y are independent.

Solution.

Scratch / Setup. Both X_2 and $Y = X_1 X_2$ take values in $\{-1, 1\}$. To prove independence, it is enough to compute

$$P(X_2 = a, Y = b)$$

for $a, b \in \{-1, 1\}$ and show it factors as $P(X_2 = a)P(Y = b)$.

Fix $a, b \in \{-1, 1\}$. Since $Y = X_1 X_2$, the event $\{X_2 = a, Y = b\}$ is the same as

$$\{X_2 = a,\ X_1 X_2 = b\} = \{X_2 = a,\ X_1 = b/a\}.$$

Therefore, by independence of X_1 and X_2,

$$P(X_2 = a, Y = b) = P(X_2 = a)P(X_1 = b/a).$$

Because $b/a \in \{-1, 1\}$ and

$$P(X_1 = 1) = P(X_1 = -1) = \frac{1}{2},$$

we get

$$P(X_2 = a, Y = b) = P(X_2 = a) \cdot \frac{1}{2}.$$

It remains to compute $P(Y = b)$. We have

$$\begin{aligned} P(Y = 1) = P(X_1 = X_2) &= P(X_1 = 1, X_2 = 1) + P(X_1 = -1, X_2 = -1) \\ &= \frac{1}{2}p + \frac{1}{2}(1 - p) = \frac{1}{2}. \end{aligned}$$

Similarly,

$$P(Y = -1) = 1 - P(Y = 1) = \frac{1}{2}.$$

Hence

$$P(X_2 = a, Y = b) = P(X_2 = a) \cdot \frac{1}{2} = P(X_2 = a)P(Y = b)$$

for all $a, b \in \{-1, 1\}$. Therefore X_2 and Y are independent.

Final: $\boxed{X_2 \text{ and } Y \text{ are independent}}$

Problem 6.28

Let X and Y be independent Geom(p) random variables. Let

$$V = \min(X, Y)$$

and

$$W = \begin{cases} 0, & \text{if } X < Y \\ 1, & \text{if } X = Y \\ 2, & \text{if } X > Y. \end{cases}$$

Find the joint probability mass function of V and W and show that V and W are independent.

Hint. Use the joint probability mass function of X and Y to compute the joint probability mass function of V and W.

Solution.

Scratch / Setup. Let $q = 1 - p$. Since $X, Y \sim \text{Geom}(p)$ independently, for $k \geq 1$,

$$P(X = k) = P(Y = k) = pq^{k-1}.$$

For $v \geq 1$:

$$P(V = v, W = 0) = P(X = v,\ Y > v),$$
$$P(V = v, W = 1) = P(X = v,\ Y = v),$$
$$P(V = v, W = 2) = P(Y = v,\ X > v).$$

For $v = 1, 2, \ldots,$

$$P(V = v, W = 0) = P(X = v)P(Y > v) = pq^{v-1} \cdot q^v = pq^{2v-1},$$

using independence and $P(Y > v) = q^v$. Similarly,

$$P(V = v, W = 2) = pq^{2v-1}.$$

Also,

$$P(V = v, W = 1) = P(X = v, Y = v) = p^2q^{2v-2}.$$

Thus the joint p.m.f. is

$$p_{V,W}(v, w) = \begin{cases} pq^{2v-1}, & v \geq 1,\ w = 0, \\ p^2q^{2v-2}, & v \geq 1,\ w = 1, \\ pq^{2v-1}, & v \geq 1,\ w = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Now compute the marginals. For $v \geq 1$,

$$p_V(v) = pq^{2v-1} + p^2q^{2v-2} + pq^{2v-1} = q^{2v-2}\big(2pq + p^2\big) = p(2-p)q^{2v-2}.$$

For W,

$$P(W=1)=\sum_{v=1}^{\infty}p^2q^{2v-2}=p^2\sum_{j=0}^{\infty}q^{2j}=\frac{p^2}{1-q^2}=\frac{p}{2-p}.$$

By symmetry, $P(W=0)=P(W=2)$, and so

$$P(W=0)=P(W=2)=\frac{1-P(W=1)}{2}=\frac{1-p}{2-p}.$$

Finally, for $v\ge 1$,

$$p_V(v)P(W=0)=p(2-p)q^{2v-2}\cdot\frac{1-p}{2-p}=pq^{2v-1}=p_{V,W}(v,0),$$

$$p_V(v)P(W=1)=p(2-p)q^{2v-2}\cdot\frac{p}{2-p}=p^2q^{2v-2}=p_{V,W}(v,1),$$

and similarly $p_V(v)P(W=2)=p_{V,W}(v,2)$. Hence V and W are independent.

Final: $\boxed{p_{V,W}(v,w)=\begin{cases}pq^{2v-1}, & v\ge 1,\ w=0,\\ p^2q^{2v-2}, & v\ge 1,\ w=1,\\ pq^{2v-1}, & v\ge 1,\ w=2,\\ 0, & \text{otherwise}\end{cases}}$

Final: $\boxed{p_V(v)=p(2-p)q^{2v-2}\text{ for }v\ge 1}$

Final: $\boxed{P(W=0)=P(W=2)=\frac{1-p}{2-p},\quad P(W=1)=\frac{p}{2-p}}$

Final: $\boxed{V\text{ and }W\text{ are independent}}$

Problem 6.30

Suppose that $X \sim \text{Geom}(p)$ and $Y \sim \text{Poisson}(\lambda)$ are independent. Find the probability $P(X-1=Y)$.

Solution.

Scratch / Setup. Since $Y \sim \text{Poisson}(\lambda)$ takes values $0, 1, 2, \ldots$, the event $\{X-1=Y\}$ is the disjoint union

$$\bigcup_{k=0}^{\infty} \{X = k+1,\ Y = k\}.$$

Use independence and the p.m.f.s

$$P(X=k+1) = p(1-p)^k, \qquad P(Y=k) = e^{-\lambda}\frac{\lambda^k}{k!}.$$

By independence,

$$P(X-1=Y) = \sum_{k=0}^{\infty} P(X=k+1)P(Y=k)$$

$$= \sum_{k=0}^{\infty} p(1-p)^k e^{-\lambda}\frac{\lambda^k}{k!} = pe^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda(1-p))^k}{k!}.$$

Recognizing the exponential series,

$$P(X-1=Y) = pe^{-\lambda}e^{\lambda(1-p)} = pe^{-\lambda p}.$$

Final: $\boxed{P(X-1=Y) = pe^{-\lambda p}}$

Problem 6.34

Let (X, Y) be a uniformly distributed random point on the quadrilateral D with vertices $(0,0)$, $(2,0)$, $(1,1)$ and $(0,1)$.

(a) Find the joint density function of (X, Y) and the marginal density functions of X and Y.

(b) Find $E[X]$ and $E[Y]$.

(c) Are X and Y independent?

Solution.

Scratch / Setup. The quadrilateral can be described as

$$D = \{(x,y) : 0 \le y \le 1,\ 0 \le x \le 2-y\}.$$

Its area is

$$\int_0^1 (2-y)\,dy = \frac{3}{2}.$$

So the joint density is constant $2/3$ on D. For marginals:

- If $0 \le x \le 1$, then y ranges from 0 to 1.
- If $1 \le x \le 2$, then y ranges from 0 to $2-x$.
- If $0 \le y \le 1$, then x ranges from 0 to $2-y$.

(a) Since (X, Y) is uniform on D,

$$f_{X,Y}(x,y) = \begin{cases} \frac{2}{3}, & (x,y) \in D, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$f_{X,Y}(x,y) = \begin{cases} \frac{2}{3}, & 0 \le y \le 1,\ 0 \le x \le 2-y, \\ 0, & \text{otherwise.} \end{cases}$$

For the marginal of X,

$$f_X(x) = \begin{cases} \int_0^1 \frac{2}{3}\,dy = \frac{2}{3}, & 0 \le x \le 1, \\ \int_0^{2-x} \frac{2}{3}\,dy = \frac{4}{3} - \frac{2}{3}x, & 1 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

For the marginal of Y,

$$f_Y(y) = \begin{cases} \int_0^{2-y} \frac{2}{3}\,dx = \frac{4}{3} - \frac{2}{3}y, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Using the marginals,

$$E[X] = \int_0^1 x \cdot \frac{2}{3}\, dx + \int_1^2 x\left(\frac{4}{3} - \frac{2}{3}x\right) dx = \frac{7}{9}.$$

Also,

$$E[Y] = \int_0^1 y\left(\frac{4}{3} - \frac{2}{3}y\right) dy = \frac{4}{9}.$$

(c) X and Y are not independent. For instance,

$$P\left(X > \frac{3}{2},\ Y > \frac{1}{2}\right) = 0$$

because every point in D satisfies $x+y \le 2$, so $x > \frac{3}{2}$ and $y > \frac{1}{2}$ cannot happen together. But

$$P\left(X > \frac{3}{2}\right) > 0, \qquad P\left(Y > \frac{1}{2}\right) > 0,$$

so the intersection probability is not the product of the marginals.

Final: $\boxed{\text{(a) } f_{X,Y}(x,y) = \tfrac{2}{3} \text{ for } (x,y) \in D,\ 0 \text{ otherwise}}$

Final: $\boxed{f_X(x) = \begin{cases} \frac{2}{3}, & 0 \le x \le 1 \\ \frac{4}{3} - \frac{2}{3}x, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}}$

Final: $\boxed{f_Y(y) = \begin{cases} \frac{4}{3} - \frac{2}{3}y, & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}}$

Final: $\boxed{\text{(b) } E[X] = \tfrac{7}{9},\ E[Y] = \tfrac{4}{9}, \qquad \text{(c) } X \text{ and } Y \text{ are not independent}}$

Problem 6.36

Suppose that X, Y are jointly continuous with joint probability density function

$$f(x,y) = ce^{-x^2/2-(x-y)^2/2}, \qquad x, y \in (-\infty, \infty),$$

for some constant c.

(a) Find the value of the constant c.

(b) Find the marginal density functions of X and Y.

(c) Determine whether X and Y are independent.

Solution.

Scratch / Setup. For normalization,

$$1 = \iint_{\mathbb{R}^2} ce^{-x^2/2-(x-y)^2/2}\,dy\,dx.$$

Integrate first in y and use the hint:

$$\int_{-\infty}^{\infty} e^{-(x-y)^2/2}\,dy = \sqrt{2\pi}.$$

For f_Y, complete the square:

$$\frac{x^2}{2} + \frac{(x-y)^2}{2} = x^2 - xy + \frac{y^2}{2} = \left(x - \frac{y}{2}\right)^2 + \frac{y^2}{4}.$$

(a) We compute

$$1 = c\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2/2-(x-y)^2/2}\,dy\,dx.$$

For fixed x,

$$\int_{-\infty}^{\infty} e^{-(x-y)^2/2}\,dy = \sqrt{2\pi},$$

so

$$1 = c\sqrt{2\pi}\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = c\sqrt{2\pi}\sqrt{2\pi} = c(2\pi).$$

Hence

$$c = \frac{1}{2\pi}.$$

(b) Using the value of c,

$$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi}e^{-x^2/2-(x-y)^2/2}\,dy = \frac{1}{2\pi}e^{-x^2/2}\int_{-\infty}^{\infty} e^{-(x-y)^2/2}\,dy$$

$$= \frac{1}{2\pi} e^{-x^2/2}\sqrt{2\pi} = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

So $X \sim N(0,1)$.

For Y,

$$f_Y(y) = \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-x^2/2-(x-y)^2/2}\, dx.$$

Complete the square:

$$\frac{x^2}{2} + \frac{(x-y)^2}{2} = \left(x - \frac{y}{2}\right)^2 + \frac{y^2}{4}.$$

Therefore,

$$f_Y(y) = \frac{1}{2\pi} e^{-y^2/4} \int_{-\infty}^{\infty} e^{-(x-y/2)^2}\, dx.$$

Since $\int_{-\infty}^{\infty} e^{-u^2}\, du = \sqrt{\pi}$,

$$f_Y(y) = \frac{1}{2\pi} e^{-y^2/4}\sqrt{\pi} = \frac{1}{2\sqrt{\pi}} e^{-y^2/4} = \frac{1}{\sqrt{4\pi}} e^{-y^2/4}.$$

Thus $Y \sim N(0,2)$.

(c) If X and Y were independent, then $f_{X,Y}(x,y) = f_X(x)f_Y(y)$. But

$$f_X(x)f_Y(y) = \frac{1}{2\pi} e^{-x^2/2-y^2/4},$$

while

$$f_{X,Y}(x,y) = \frac{1}{2\pi} e^{-x^2/2-(x-y)^2/2}.$$

These are not equal in general because the exponent in $f_{X,Y}$ contains the cross term xy. Therefore X and Y are not independent.

Final: $\boxed{\text{(a) } c = \frac{1}{2\pi}}$

Final: $\boxed{\text{(b) } f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad f_Y(y) = \frac{1}{\sqrt{4\pi}} e^{-y^2/4}}$

Final: $\boxed{\text{(c) } X \text{ and } Y \text{ are not independent}}$

Homework 10

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: April 17, 2026

Contents

Problem 7.2

Let X and Y be independent Bernoulli random variables with parameters p and r, respectively. Find the distribution of $X + Y$.

Solution.

Scratch / Setup. Let
$$S = X + Y.$$
Since $X, Y \in \{0, 1\}$, the only possible values of S are $0, 1, 2$.

The relevant combinations are:
$$\begin{aligned} S = 0 &\iff (X, Y) = (0, 0), \\ S = 1 &\iff (X, Y) \in \{(1, 0), (0, 1)\}, \\ S = 2 &\iff (X, Y) = (1, 1). \end{aligned}$$
Because X and Y are independent Bernoulli random variables,
$$P(X = 1) = p, \quad P(X = 0) = 1 - p, \quad P(Y = 1) = r, \quad P(Y = 0) = 1 - r.$$

We are computing the probability mass function of $S = X + Y$.

Since S can take only the values $0, 1, 2$, we compute these probabilities directly. First,
$$P(S = 0) = P(X = 0, Y = 0) = P(X = 0)P(Y = 0) = (1 - p)(1 - r).$$
Next,
$$P(S = 2) = P(X = 1, Y = 1) = P(X = 1)P(Y = 1) = pr.$$
Finally,
$$\begin{aligned} P(S = 1) &= P(X = 1, Y = 0) + P(X = 0, Y = 1) \\ &= P(X = 1)P(Y = 0) + P(X = 0)P(Y = 1) \\ &= p(1 - r) + (1 - p)r \\ &= p + r - 2pr. \end{aligned}$$
Therefore the pmf of $X + Y$ is
$$p_S(s) = P(S = s) = \begin{cases} (1 - p)(1 - r), & s = 0, \\ p + r - 2pr, & s = 1, \\ pr, & s = 2, \\ 0, & \text{otherwise.} \end{cases}$$
This is a binomial distribution only in the special case $p = r$.

Final: $\boxed{P(X + Y = 0) = (1 - p)(1 - r), \quad P(X + Y = 1) = p + r - 2pr}$

Final: $\boxed{P(X + Y = 2) = pr}$

Problem 7.16

Let X be a Poisson random variable with parameter λ, and Y an independent Bernoulli random variable with parameter p. Find the probability mass function of $X+Y$.

Solution.

Scratch / Setup. Let
$$S = X + Y.$$
Since $X \in \{0,1,2,\dots\}$ and $Y \in \{0,1\}$, the support of S is also
$$\{0,1,2,\dots\}.$$
The cleanest computation is to condition on Y:
$$P(S=n) = P(Y=0)P(X=n) + P(Y=1)P(X=n-1),$$
with the convention that $P(X=-1)=0$.
Also,
$$P(X=k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0,1,2,\dots$$
and
$$P(Y=0) = 1-p, \qquad P(Y=1) = p.$$

We want the probability mass function of $S = X+Y$.
For $n \ge 0$,
$$P(S=n) = P(Y=0)P(X=n) + P(Y=1)P(X=n-1).$$
If $n=0$, the second term is zero, so
$$P(S=0) = (1-p)P(X=0) = (1-p)e^{-\lambda}.$$
If $n \ge 1$, then
$$\begin{aligned} P(S=n) &= (1-p)e^{-\lambda}\frac{\lambda^n}{n!} + p\,e^{-\lambda}\frac{\lambda^{n-1}}{(n-1)!} \\ &= e^{-\lambda}\frac{\lambda^{n-1}}{n!}\big((1-p)\lambda + np\big). \end{aligned}$$
Hence the pmf of $X+Y$ is
$$p_{X+Y}(n) = \begin{cases} (1-p)e^{-\lambda}, & n = 0, \\ e^{-\lambda}\dfrac{\lambda^{n-1}}{n!}\big((1-p)\lambda + np\big), & n = 1,2,3,\dots \end{cases}$$

Final: $\boxed{p_{X+Y}(0) = (1-p)e^{-\lambda}, \quad p_{X+Y}(n) = e^{-\lambda}\frac{\lambda^{n-1}}{n!}\big((1-p)\lambda+np\big) \text{ for } n \ge 1}$

Problem 7.20

Let X have density $f_X(x) = 2x$ for $0 < x < 1$ and 0 outside of $(0, 1)$, and let Y be uniform on the interval $(1, 2)$. Assume X and Y independent.

(b) Find the density function of $X + Y$.

Solution.

Scratch / Setup. Let

$$T = X + Y.$$

Since $X \in (0, 1)$ and $Y \in (1, 2)$, we must have

$$T \in (1, 3).$$

By convolution,

$$f_T(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x)\, dx.$$

Here

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise}, \end{cases} \qquad f_Y(y) = \begin{cases} 1, & 1 < y < 2, \\ 0, & \text{otherwise}. \end{cases}$$

The integrand is nonzero exactly when

$$0 < x < 1 \quad \text{and} \quad 1 < t - x < 2,$$

that is,

$$\max(0, t - 2) < x < \min(1, t - 1).$$

This leads to two cases:

$$1 < t < 2 \quad \text{and} \quad 2 \le t < 3.$$

We are finding the density of $T = X + Y$.

For any real t,

$$f_T(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x)\, dx.$$

If $t \notin (1, 3)$, there is no overlap of the supports, so $f_T(t) = 0$.

If $1 < t < 2$, then $\max(0, t - 2) = 0$ and $\min(1, t - 1) = t - 1$, so

$$f_T(t) = \int_0^{t-1} 2x\, dx = (t - 1)^2.$$

If $2 \le t < 3$, then $\max(0, t - 2) = t - 2$ and $\min(1, t - 1) = 1$, so

$$f_T(t) = \int_{t-2}^{1} 2x\, dx = 1 - (t - 2)^2.$$

Therefore

$$f_{X+Y}(t) = \begin{cases} (t-1)^2, & 1 < t < 2, \\ 1-(t-2)^2, & 2 \le t < 3, \\ 0, & \text{otherwise.} \end{cases}$$

As a quick check,

$$\int_1^2 (t-1)^2\,dt + \int_2^3 \left(1-(t-2)^2\right) dt = \frac{1}{3} + \frac{2}{3} = 1,$$

so this is a valid density.

Final: $\boxed{f_{X+Y}(t) = \begin{cases} (t-1)^2, & 1 < t < 2 \\ 1-(t-2)^2, & 2 \le t < 3 \\ 0, & \text{otherwise} \end{cases}}$

Problem 7.24

Suppose that X, Y, and Z are independent mean zero normal random variables. Find $P(X + 2Y - 3Z > 0)$.

Solution.

Scratch / Setup. Let
$$W = X + 2Y - 3Z.$$
Because X, Y, Z are independent normal random variables, any linear combination of them is also normal. So W is normal.

Its mean is
$$E[W] = E[X] + 2E[Y] - 3E[Z] = 0.$$
If we write
$$\mathrm{Var}(X) = \sigma_X^2, \quad \mathrm{Var}(Y) = \sigma_Y^2, \quad \mathrm{Var}(Z) = \sigma_Z^2,$$
then independence gives
$$\mathrm{Var}(W) = \sigma_X^2 + 4\sigma_Y^2 + 9\sigma_Z^2.$$
So W is a mean-zero normal random variable, hence its distribution is symmetric about 0.

We are asked to compute the probability
$$P(X + 2Y - 3Z > 0) = P(W > 0),$$
where
$$W = X + 2Y - 3Z.$$
Since W is a linear combination of independent normal random variables, it is itself normal. Its mean is
$$E[W] = 0,$$
and its variance is
$$\mathrm{Var}(W) = \mathrm{Var}(X) + 4\,\mathrm{Var}(Y) + 9\,\mathrm{Var}(Z).$$
Therefore W is a centered normal random variable. A mean-zero normal distribution is symmetric about 0, so
$$P(W > 0) = \frac{1}{2}.$$

Final: $\boxed{P(X + 2Y - 3Z > 0) = \frac{1}{2}}$

Problem 7.28

Let X_1, X_2, X_3 be independent $\text{Exp}(\lambda)$ distributed random variables. Find the probability that $X_1 < X_2 < X_3$.

Solution.

Scratch / Setup. The variables X_1, X_2, X_3 are i.i.d. and continuous.

Because they are continuous,

$$P(X_i = X_j) = 0 \qquad \text{for } i \neq j,$$

so ties occur with probability 0.

There are $3! = 6$ strict orderings:

$$\begin{array}{ccc} X_1 < X_2 < X_3, & X_1 < X_3 < X_2, & X_2 < X_1 < X_3, \\ X_2 < X_3 < X_1, & X_3 < X_1 < X_2, & X_3 < X_2 < X_1. \end{array}$$

By exchangeability of i.i.d. random variables, these six orderings are equally likely.

Since X_1, X_2, X_3 are independent and identically distributed continuous random variables, each strict ordering of the three variables has the same probability. Also, because the distribution is continuous, ties have probability zero. Hence the six events

$$\begin{array}{ccc} X_1 < X_2 < X_3, & X_1 < X_3 < X_2, & X_2 < X_1 < X_3, \\ X_2 < X_3 < X_1, & X_3 < X_1 < X_2, & X_3 < X_2 < X_1 \end{array}$$

are disjoint, equally likely, and their union has probability 1.

Therefore each one has probability

$$\frac{1}{6}.$$

Final: $\boxed{P(X_1 < X_2 < X_3) = \frac{1}{6}}$

Problem 7.30

We deal five cards, one by one, from a standard deck of 52. (Dealing cards from a deck means sampling without replacement.)

(a) Find the probability that the second card is an ace and the fourth card is a king.

(b) Find the probability that the first and the fifth cards are both spades.

(c) Find the conditional probability that the second card is a king given that the last two cards are both aces.

Solution.

Scratch / Setup. We deal five cards from a uniformly shuffled deck, so all ordered 5-card sequences are equally likely.

Useful facts:

- There are 4 aces, 4 kings, and 13 spades.
- By symmetry/exchangeability, specified positions behave like any other specified positions.
- Conditioning on the last two cards being aces leaves 50 cards for the first three positions, still including all 4 kings.

We compute each part separately.

(a) We need the probability that the second card is an ace and the fourth card is a king.

By exchangeability of the positions in a random ordering of the deck, this is the same as the probability that the first card is an ace and the second card is a king. Therefore

$$P(\text{2nd ace and 4th king}) = \frac{4}{52} \cdot \frac{4}{51} = \frac{16}{2652} = \frac{4}{663}.$$

(b) We need the probability that the first and fifth cards are both spades.

Again by exchangeability, this is the same as the probability that the first two cards are both spades. Thus

$$P(\text{1st and 5th spades}) = \frac{13}{52} \cdot \frac{12}{51} = \frac{1}{17}.$$

(c) We need

$$P(\text{2nd card is a king} \mid \text{last two cards are both aces}).$$

After conditioning on the last two cards being aces, the first three positions are filled by an ordered sample from the remaining 50 cards. None of the aces is a king, so all 4 kings are still among those 50 cards. By symmetry, each of the first three positions is equally likely to contain any one of the remaining 50 cards. Hence

$$P(\text{2nd card is a king} \mid \text{last two cards are both aces}) = \frac{4}{50} = \frac{2}{25}.$$

Final: $\boxed{\text{(a) } \frac{4}{663}}$

Final: $\boxed{\text{(b) } \frac{1}{17}}$

Final: $\boxed{\text{(c) } \frac{2}{25}}$

Homework 11

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: April 24, 2026

Contents

Problem 8.6

Exercise 8.6. Let X and Y be as in Exercise 8.1. Assume additionally that X and Y are independent. Evaluate all the expectations in part (d) of Exercise 8.1.

> **Exercise 8.1.** Let n be a positive integer and p and r two real numbers in $(0, 1)$. Two random variables X and Y are defined on the same sample space. All we know about them is that $X \sim \text{Geom}(p)$ and $Y \sim \text{Bin}(n, r)$. For each expectation in parts (a)–(d) below, decide whether it can be calculated with this information, and if it can, give its value.
>
> (a) $E[X + Y]$
>
> (b) $E[XY]$
>
> (c) $E[X^2 + Y^2]$
>
> (d) $E[(X + Y)^2]$

Solution.

Scratch / Setup. We only need the expectation from part (d) of Exercise 8.1:

$$E[(X + Y)^2].$$

Expand:

$$E[(X + Y)^2] = E[X^2] + 2E[XY] + E[Y^2].$$

Because X and Y are independent,

$$E[XY] = E[X]E[Y].$$

For $X \sim \text{Geom}(p)$ on $\{1, 2, \dots\}$,

$$E[X] = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2},$$

so

$$E[X^2] = \text{Var}(X) + (E[X])^2 = \frac{1-p}{p^2} + \frac{1}{p^2} = \frac{2-p}{p^2}.$$

For $Y \sim \text{Bin}(n, r)$,

$$E[Y] = nr, \qquad \text{Var}(Y) = nr(1 - r),$$

so

$$E[Y^2] = \text{Var}(Y) + (E[Y])^2 = nr(1 - r) + n^2r^2.$$

We are computing

$$E[(X + Y)^2].$$

Using the identity

$$(X+Y)^2 = X^2 + 2XY + Y^2,$$

we get

$$E[(X+Y)^2] = E[X^2] + 2E[XY] + E[Y^2].$$

Since X and Y are independent,

$$E[XY] = E[X]E[Y] = \frac{1}{p} \cdot nr = \frac{nr}{p}.$$

Also,

$$E[X^2] = \frac{2-p}{p^2}, \qquad E[Y^2] = nr(1-r) + n^2r^2.$$

Therefore

$$E[(X+Y)^2] = \frac{2-p}{p^2} + 2\frac{nr}{p} + nr(1-r) + n^2r^2.$$

Final: $\boxed{E[(X+Y)^2] = \frac{2-p}{p^2} + \frac{2nr}{p} + nr(1-r) + n^2r^2}$

Problem 8.12

Exercise 8.12.

(a) Let Z be Gamma$(2, \lambda)$ distributed. That is, Z has the density function

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Use the definition of the moment generating function to calculate $M_Z(t) = E[e^{tZ}]$.

(b) Let X and Y be two independent Exp(λ) random variables. Recall the moment generating function of X and Y from Example 5.6 in Section 5.1. Using the approach from Section 8.3 show that Z and $X + Y$ have the same distribution.

Solution.

Scratch / Setup. For part (a), use the definition of the mgf:

$$M_Z(t) = E[e^{tZ}] = \int_0^\infty e^{tz} \lambda^2 z e^{-\lambda z}\, dz = \lambda^2 \int_0^\infty z e^{-(\lambda - t)z}\, dz.$$

This only makes sense for $t < \lambda$.

Now factor out $(\lambda - t)^{-2}$:

$$M_Z(t) = \frac{\lambda^2}{(\lambda - t)^2} \int_0^\infty (\lambda - t)^2 z e^{-(\lambda - t)z}\, dz.$$

The integrand is the Gamma$(2, \lambda - t)$ density, so the integral is 1.

For part (b), if $X, Y \sim$ Exp(λ) are independent, then

$$M_X(t) = M_Y(t) = \frac{\lambda}{\lambda - t}, \qquad t < \lambda,$$

and hence

$$M_{X+Y}(t) = M_X(t) M_Y(t) = \left(\frac{\lambda}{\lambda - t}\right)^2.$$

(a) We compute the mgf directly:

$$M_Z(t) = \int_0^\infty e^{tz} \lambda^2 z e^{-\lambda z}\, dz = \lambda^2 \int_0^\infty z e^{-(\lambda - t)z}\, dz.$$

For $t < \lambda$, this becomes

$$M_Z(t) = \frac{\lambda^2}{(\lambda - t)^2} \int_0^\infty (\lambda - t)^2 z e^{-(\lambda - t)z}\, dz.$$

The integral is 1 because it is the total integral of a Gamma$(2, \lambda - t)$ density. Therefore

$$M_Z(t) = \left(\frac{\lambda}{\lambda - t}\right)^2, \qquad t < \lambda.$$

(b) Since X and Y are independent $\mathrm{Exp}(\lambda)$ random variables,

$$M_X(t) = M_Y(t) = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

Hence

$$M_{X+Y}(t) = M_X(t)M_Y(t) = \left(\frac{\lambda}{\lambda - t}\right)^2.$$

Part (a) showed that this is exactly $M_Z(t)$ for $t < \lambda$. Since the moment generating functions agree on an interval containing 0, the distributions are the same. Thus

$$X + Y \sim \Gamma(2, \lambda).$$

Final: $\boxed{\text{(a) } M_Z(t) = \left(\frac{\lambda}{\lambda-t}\right)^2 \text{ for } t < \lambda}$

Final: $\boxed{\text{(b) } X + Y \sim \Gamma(2, \lambda)}$

Problem 8.22

Exercise 8.22. In a lottery 5 numbers are chosen from the set $\{1, \ldots, 90\}$ without replacement. We order the five numbers in increasing order and we denote by X the number of times the difference between two neighboring numbers is 1. (For example, for $\{1, 2, 3, 4, 5\}$ we have $X = 4$, for $\{3, 6, 8, 9, 24\}$ we have $X = 1$ and for $\{12, 14, 43, 75, 88\}$ we have $X = 0$.) Find $E[X]$.

Solution.

Scratch / Setup. It is cleaner to index the possible consecutive pairs of numbers rather than the four gaps in the ordered sample.

For $k = 1, \ldots, 89$, let

$$J_k = \mathbf{1}\{\text{both } k \text{ and } k+1 \text{ are chosen}\}.$$

Then every selected consecutive pair contributes exactly 1 to X, so

$$X = \sum_{k=1}^{89} J_k.$$

For a fixed k,

$$E[J_k] = P(J_k = 1) = \frac{\binom{88}{3}}{\binom{90}{5}},$$

since once k and $k+1$ are included, the remaining 3 chosen numbers can be any 3 of the other 88 numbers.

Let

$$J_k = \mathbf{1}\{\text{both } k \text{ and } k+1 \text{ are chosen}\}, \qquad k = 1, \ldots, 89.$$

Then

$$X = \sum_{k=1}^{89} J_k,$$

because X counts exactly how many consecutive integer pairs occur in the chosen 5-element set.

By linearity of expectation,

$$E[X] = \sum_{k=1}^{89} E[J_k] = \sum_{k=1}^{89} P(J_k = 1).$$

For a fixed k,

$$P(J_k = 1) = \frac{\binom{88}{3}}{\binom{90}{5}}.$$

Therefore

$$E[X] = 89\frac{\binom{88}{3}}{\binom{90}{5}}.$$

Now
$$\frac{\binom{88}{3}}{\binom{90}{5}} = \frac{88!}{3!\,85!} \cdot \frac{5!\,85!}{90!} = \frac{20}{90 \cdot 89} = \frac{2}{801}.$$
Hence
$$E[X] = 89 \cdot \frac{2}{801} = \frac{2}{9}.$$

Final: $\boxed{E[X] = \frac{2}{9}}$

Problem 8.23

Exercise 8.23. We have an urn with 20 red and 30 green balls. We draw all the balls one by one and record the colors we see. (That is, we take a sample of 50 without replacement.)

(a) Find the probability that the 28th and 29th balls are of different color.

(b) Let X be the number of times two consecutive balls are of different color. Find $E[X]$.

Solution.

Scratch / Setup. For part (a), by exchangeability any adjacent pair of positions has the same law as the first two draws.

For part (b), let

$$I_k = \mathbf{1}\{\text{balls } k \text{ and } k+1 \text{ have different colors}\}, \qquad k = 1, \dots, 49.$$

Then

$$X = \sum_{k=1}^{49} I_k.$$

So

$$E[X] = \sum_{k=1}^{49} E[I_k] = 49\, P(\text{two consecutive balls are different colors}).$$

(a) By exchangeability, the colors of the 28th and 29th draws have the same joint distribution as the colors of the first two draws. So

$$\begin{aligned} P(\text{28th and 29th different}) &= P(RG) + P(GR) \\ &= \frac{20}{50} \cdot \frac{30}{49} + \frac{30}{50} \cdot \frac{20}{49} \\ &= \frac{24}{49}. \end{aligned}$$

(b) Let

$$I_k = \mathbf{1}\{\text{balls } k \text{ and } k+1 \text{ have different colors}\}, \qquad k = 1, \dots, 49.$$

Then

$$X = \sum_{k=1}^{49} I_k.$$

By linearity of expectation,

$$E[X] = \sum_{k=1}^{49} E[I_k] = \sum_{k=1}^{49} P(I_k = 1).$$

Again by exchangeability, $P(I_k = 1)$ is the same for every k, and part (a) shows that this common value is $24/49$. Therefore

$$E[X] = 49 \cdot \frac{24}{49} = 24.$$

Final: $\boxed{\text{(a) } \frac{24}{49}}$

Final: $\boxed{\text{(b) } E[X] = 24}$

Problem 8.36

Exercise 8.36. I roll a fair die four times. Let X be the number of different outcomes that I see. (For example, if the die rolls are $5, 3, 6, 6$ then $X = 3$ because the different outcomes are 3, 5 and 6.)

(a) Find the mean of X.

(b) Find the variance of X.

Solution.

Scratch / Setup. For $j = 1, \dots, 6$, let

$$I_j = \mathbf{1}\{\text{face } j \text{ appears at least once in the four rolls}\}.$$

Then

$$X = I_1 + \cdots + I_6.$$

For a fixed face j,

$$P(I_j = 1) = 1 - P(\text{face } j \text{ never appears}) = 1 - \left(\frac{5}{6}\right)^4.$$

Call this probability p.

For $i \neq j$,

$$P(I_i = 1, I_j = 1) = 1 - P(i \text{ absent}) - P(j \text{ absent}) + P(i, j \text{ both absent}),$$

so

$$P(I_i = 1, I_j = 1) = 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4.$$

Call this probability q.

Then

$$\operatorname{Var}(X) = \sum_{j=1}^{6} \operatorname{Var}(I_j) + 2 \sum_{1 \le i < j \le 6} \operatorname{Cov}(I_i, I_j).$$

(a) Let

$$I_j = \mathbf{1}\{\text{face } j \text{ appears at least once}\}, \qquad j = 1, \dots, 6.$$

Then

$$X = I_1 + \cdots + I_6.$$

By linearity of expectation,

$$E[X] = \sum_{j=1}^{6} E[I_j] = \sum_{j=1}^{6} P(I_j = 1).$$

For any fixed j,

$$P(I_j = 1) = 1 - \left(\frac{5}{6}\right)^4.$$

Therefore

$$E[X] = 6\left(1 - \left(\frac{5}{6}\right)^4\right) = \frac{671}{216}.$$

(b) Write

$$p = P(I_1 = 1) = 1 - \left(\frac{5}{6}\right)^4 = \frac{671}{1296}.$$

Then

$$\mathrm{Var}(I_j) = p(1-p).$$

For two distinct faces $i \neq j$,

$$\begin{aligned} q &= P(I_i = 1, I_j = 1) \\ &= 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4 = \frac{151}{648}, \end{aligned}$$

so

$$\mathrm{Cov}(I_i, I_j) = q - p^2.$$

Since there are 6 diagonal terms and $\binom{6}{2} = 15$ distinct pairs,

$$\mathrm{Var}(X) = 6p(1-p) + 2\binom{6}{2}(q - p^2).$$

Therefore

$$\mathrm{Var}(X) = 6p(1-p) + 30(q - p^2) = \frac{20855}{46656}.$$

Final: $\boxed{\text{(a) } E[X] = 6\left(1 - \left(\tfrac{5}{6}\right)^4\right) = \tfrac{671}{216}}$

Final: $\boxed{\text{(b) } \mathrm{Var}(X) = \tfrac{20855}{46656}}$

Problem 8.40

Exercise 8.40. In a certain lottery 5 numbers are drawn without replacement out of the set $\{1, 2, \ldots, 90\}$ randomly each week. Let X be the number of different numbers drawn during the first four weeks of the year. (For example, if the draws are $\{1, 2, 3, 4, 5\}$, $\{2, 3, 4, 5, 6\}$, $\{3, 4, 5, 6, 7\}$ and $\{4, 5, 6, 7, 8\}$ then $X = 8$.)

(a) Find $E[X]$.

Solution.

Scratch / Setup. For each number $m \in \{1, \ldots, 90\}$, let

$$I_m = \mathbf{1}\{\text{number } m \text{ appears at least once in the first four weeks}\}.$$

Then

$$X = \sum_{m=1}^{90} I_m.$$

So

$$E[X] = \sum_{m=1}^{90} E[I_m] = 90\, P(\text{a fixed number appears at least once}).$$

For a fixed number m, in one week

$$P(m \text{ is not drawn}) = \frac{\binom{89}{5}}{\binom{90}{5}} = \frac{17}{18}.$$

Since the weekly drawings are independent, the probability m is never drawn in the first four weeks is

$$\left(\frac{17}{18}\right)^4.$$

For each $m \in \{1, \ldots, 90\}$, let

$$I_m = \mathbf{1}\{\text{number } m \text{ is drawn at least once during the first four weeks}\}.$$

Then

$$X = \sum_{m=1}^{90} I_m.$$

By linearity of expectation,

$$E[X] = \sum_{m=1}^{90} E[I_m] = 90\, P(I_1 = 1).$$

Now fix a number, say 1. In one week,

$$P(1 \text{ is not drawn}) = \frac{\binom{89}{5}}{\binom{90}{5}} = \frac{17}{18}.$$

Because the four weekly drawings are independent,

$$P(1 \text{ is never drawn in the first four weeks}) = \left(\frac{17}{18}\right)^4.$$

Hence

$$P(I_1 = 1) = 1 - \left(\frac{17}{18}\right)^4.$$

Therefore

$$E[X] = 90\left(1 - \left(\frac{17}{18}\right)^4\right).$$

Final: $\boxed{E[X] = 90\left(1 - \left(\tfrac{17}{18}\right)^4\right)}$

Problem 8.42

Exercise 8.42. The random variables $X_1, \ldots, X_n$ are i.i.d. We also know that $E[X_1] = 0$, $E[X_1^2] = a$, $E[X_1^3] = b$ and $E[X_1^4] = c$. Let $\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$. Find the fourth moment of $\bar{X}_n$.

Solution.

Scratch / Setup. Let

$$S_n = X_1 + \cdots + X_n.$$

Then

$$\bar{X}_n = \frac{S_n}{n}, \qquad E[\bar{X}_n^4] = \frac{1}{n^4} E[S_n^4].$$

Expand S_n^4 by grouping terms according to exponent pattern:

$$S_n^4 = \sum_{i=1}^{n} X_i^4 + 4 \sum_{i \neq j} X_i^3 X_j + 6 \sum_{i<j} X_i^2 X_j^2 + 12 \sum_{\substack{i,j,k \\ \text{distinct}}} X_i^2 X_j X_k + 24 \sum_{i<j<k<\ell} X_i X_j X_k X_\ell.$$

Because the X_i are independent and $E[X_i] = 0$, any term containing a factor with odd first power has expectation 0. So only

$$\sum X_i^4 \quad \text{and} \quad \sum_{i<j} X_i^2 X_j^2$$

survive after taking expectation.

Let

$$S_n = X_1 + \cdots + X_n.$$

Then

$$E[\bar{X}_n^4] = \frac{1}{n^4} E[S_n^4].$$

Expanding S_n^4 and taking expectations, all terms with an odd factor vanish because the X_i are independent and $E[X_i] = 0$. Thus only two types of terms remain:

$$E[S_n^4] = \sum_{i=1}^{n} E[X_i^4] + 6 \sum_{i<j} E[X_i^2 X_j^2].$$

By identical distribution,

$$E[X_i^4] = c$$

for every i, and by independence,

$$E[X_i^2 X_j^2] = E[X_i^2] E[X_j^2] = a^2 \qquad (i \neq j).$$

Therefore

$$E[S_n^4] = nc + 6 \binom{n}{2} a^2 = nc + 3n(n-1)a^2.$$

Dividing by n^4 gives

$$E[\bar{X}_n^4] = \frac{nc + 3n(n-1)a^2}{n^4} = \frac{c + 3(n-1)a^2}{n^3}.$$

Final: $\boxed{E[\bar{X}_n^4] = \frac{c+3(n-1)a^2}{n^3}}$

Problem 8.48

Exercise 8.48. Suppose there are 100 cards numbered 1 through 100. Draw a card at random. Let X be the number of digits on the card (between 1 and 3) and let Y be the number of zeros. Find $\mathrm{Cov}(X, Y)$.

Solution.

Scratch / Setup. Partition the numbers $1, \dots, 100$ into:

- 1-digit numbers: $1, \dots, 9$ (9 numbers),
- 2-digit numbers: $10, \dots, 99$ (90 numbers),
- the 3-digit number: 100.

For Y:

- among $10, \dots, 99$, only $10, 20, \dots, 90$ contain a zero, each with one zero,
- 100 has two zeros.

So we can compute $E[X]$, $E[Y]$, and $E[XY]$ by averaging over these cases, then use

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

We compute

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

First,

$$E[X] = \frac{1}{100}\big(9 \cdot 1 + 90 \cdot 2 + 1 \cdot 3\big) = \frac{192}{100} = \frac{48}{25}.$$

Next, the total number of zeros among the integers $1, \dots, 100$ is

$$9 \cdot 1 + 2 = 11,$$

since $10, 20, \dots, 90$ contribute one zero each and 100 contributes two zeros. Hence

$$E[Y] = \frac{11}{100}.$$

For $E[XY]$, the only numbers with $Y > 0$ are $10, 20, \dots, 90$ and 100. Thus

$$E[XY] = \frac{1}{100}\big(9 \cdot (2 \cdot 1) + 1 \cdot (3 \cdot 2)\big) = \frac{18 + 6}{100} = \frac{24}{100} = \frac{6}{25}.$$

Therefore

$$\mathrm{Cov}(X, Y) = \frac{6}{25} - \frac{48}{25} \cdot \frac{11}{100} = \frac{600 - 528}{2500} = \frac{72}{2500} = \frac{18}{625}.$$

Final: $\boxed{\mathrm{Cov}(X, Y) = \frac{18}{625}}$

Problem 8.54

Exercise 8.54. Suppose that for the random variables X, Y we have $E[X] = 2$, $E[Y] = 1$, $E[X^2] = 5$, $E[Y^2] = 10$ and $E[XY] = 1$.

(a) Compute $\mathrm{Corr}(X, Y)$.

(b) Find a number c so that X and $X + cY$ are uncorrelated.

Solution.

Scratch / Setup. We are given

$$E[X] = 2, \quad E[Y] = 1, \quad E[X^2] = 5, \quad E[Y^2] = 10, \quad E[XY] = 1.$$

So

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1 - 2 \cdot 1 = -1.$$

Also,

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = 5 - 4 = 1,$$
$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2 = 10 - 1 = 9.$$

For part (b),

$$\mathrm{Cov}(X, X + cY) = \mathrm{Cov}(X, X) + c\,\mathrm{Cov}(X, Y) = \mathrm{Var}(X) + c\,\mathrm{Cov}(X, Y).$$

Set this equal to 0 and solve for c.

(a) We first compute the covariance:

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1 - 2 \cdot 1 = -1.$$

Next,

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = 5 - 4 = 1,$$

and

$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2 = 10 - 1 = 9.$$

Therefore

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = \frac{-1}{\sqrt{1 \cdot 9}} = -\frac{1}{3}.$$

(b) We want X and $X + cY$ to be uncorrelated, so we set

$$\mathrm{Cov}(X, X + cY) = 0.$$

By linearity of covariance,

$$\mathrm{Cov}(X, X + cY) = \mathrm{Cov}(X, X) + c\,\mathrm{Cov}(X, Y) = \mathrm{Var}(X) + c\,\mathrm{Cov}(X, Y).$$

Using the values from part (a),

$$0 = 1 + c(-1) = 1 - c,$$

so

$$c = 1.$$

Final: $\boxed{\text{(a) } \operatorname{Corr}(X,Y) = -\frac{1}{3}}$

Final: $\boxed{\text{(b) } c = 1}$

Homework 12

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MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

Due: May 1, 2026

Contents

Problem 9.2

Let X be an exponential random variable with parameter $\lambda = \frac{1}{2}$.

(a) Use Markov's inequality to find an upper bound for $P(X > 6)$.

(b) Use Chebyshev's inequality to find an upper bound for $P(X > 6)$.

(c) Explicitly compute the probability above and compare with the upper bounds you derived.

Solution.

Scratch / Setup. For $X \sim \text{Exp}(\lambda)$ with $\lambda = \frac{1}{2}$,

$$E[X] = \frac{1}{\lambda} = 2, \qquad \text{Var}(X) = \frac{1}{\lambda^2} = 4.$$

Relevant facts:

- $X \geq 0$ almost surely, so Markov's inequality applies.
- Chebyshev's inequality says

$$P\big(|X - E[X]| \geq t\big) \leq \frac{\text{Var}(X)}{t^2}.$$

- Since $E[X] = 2$, the event $\{X > 6\}$ implies $\{|X - 2| \geq 4\}$.
- For an exponential random variable,

$$P(X > x) = e^{-\lambda x}, \qquad x \geq 0.$$

(a) Since $X \geq 0$, Markov's inequality gives

$$P(X > 6) \leq \frac{E[X]}{6} = \frac{2}{6} = \frac{1}{3}.$$

Final: $\boxed{P(X > 6) \leq \frac{1}{3}}$

(b) Because $\{X > 6\} \subseteq \{|X - 2| \geq 4\}$, we have

$$P(X > 6) \leq P(|X - 2| \geq 4).$$

Now apply Chebyshev's inequality:

$$P(|X - 2| \geq 4) \leq \frac{\text{Var}(X)}{4^2} = \frac{4}{16} = \frac{1}{4}.$$

Therefore

$$P(X > 6) \leq \frac{1}{4}.$$

Final: $\boxed{P(X > 6) \leq \frac{1}{4}}$

(c) The exponential tail can be computed exactly:

$$P(X > 6) = e^{-\lambda \cdot 6} = e^{-3} \approx 0.0498.$$

Comparing the three values,

$$e^{-3} \approx 0.0498 < \frac{1}{4} < \frac{1}{3}.$$

So both inequalities give valid upper bounds, and the Chebyshev bound is better than the Markov bound because $\frac{1}{4}$ is smaller and closer to the exact probability.

Final: $\boxed{P(X > 6) = e^{-3} \approx 0.0498, \quad \text{and Chebyshev gives the better upper bound}}$

Problem 9.6

Nate is a competitive eater specializing in eating hot dogs. From past experience we know that it takes him on average 15 seconds to consume one hot dog, with a standard deviation of 4 seconds. In this year's hot dog eating contest he hopes to consume 64 hot dogs in just 15 minutes. Use the CLT to approximate the probability that he achieves this feat of skill.

Solution.

Scratch / Setup. Let X_i be the time, in seconds, that Nate needs to consume the ith hot dog, and let

$$S_{64} = X_1 + \cdots + X_{64}.$$

Then Nate succeeds exactly when $S_{64} \le 15$ minutes $= 900$ seconds.

From the data,

$$E[X_i] = 15, \qquad \mathrm{SD}(X_i) = 4, \qquad \mathrm{Var}(X_i) = 16.$$

Assuming the X_i are i.i.d.,

$$E[S_{64}] = 64 \cdot 15 = 960, \qquad \mathrm{Var}(S_{64}) = 64 \cdot 16 = 1024, \qquad \mathrm{SD}(S_{64}) = 32.$$

Use the CLT because the exact distribution of S_{64} is not given:

$$\frac{S_{64} - 960}{32} \approx N(0,1).$$

We want the probability that Nate consumes 64 hot dogs within 15 minutes, so

$$P(\text{Nate succeeds}) = P(S_{64} \le 900).$$

By the central limit theorem,

$$P(S_{64} \le 900) \approx P\left(\frac{S_{64} - 960}{32} \le \frac{900 - 960}{32}\right) = \Phi\left(\frac{-60}{32}\right) = \Phi(-1.875).$$

Therefore,

$$P(S_{64} \le 900) \approx \Phi(-1.875) \approx 0.0304.$$

So the CLT approximation says that Nate has about a 3.04% chance of finishing 64 hot dogs in 15 minutes.

Final: $\boxed{P(S_{64} \le 900) \approx \Phi(-1.875) \approx 0.0304}$

Problem 9.18

Suppose that $X \sim \text{NegBin}(100, \frac{1}{3})$.

(a) Give an upper bound for $P(X > 500)$ using Markov's inequality.

(b) Give an upper bound for $P(X > 500)$ using Chebyshev's inequality.

(c) Recall that X can be written as the sum of 100 i.i.d. random variables with $\text{Geom}(\frac{1}{3})$ distribution. Use this representation and the CLT to estimate $P(X > 500)$.

(d) Recall that X can be represented as the number of trials needed to get the 100th success in a sequence of i.i.d. trials of success probability $\frac{1}{3}$. Estimate $P(X > 500)$ by looking at the distribution of the number of successes within the first 500 trials.

Solution.

Scratch / Setup. Here $X \sim \text{NegBin}(100, \frac{1}{3})$ means that X is the number of trials needed to obtain the 100th success when each trial succeeds with probability

$$p = \frac{1}{3}.$$

With $r = 100$ and this convention,

$$E[X] = \frac{r}{p} = \frac{100}{1/3} = 300,$$

$$\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{100(2/3)}{(1/3)^2} = 600, \qquad \text{SD}(X) = \sqrt{600} = 10\sqrt{6}.$$

For part (c), write

$$X = G_1 + \cdots + G_{100},$$

where the G_i are i.i.d. $\text{Geom}(\frac{1}{3})$ random variables.

For part (d), let N be the number of successes in the first 500 trials. Then

$$N \sim \text{Bin}\left(500, \frac{1}{3}\right), \qquad \{X > 500\} = \{N \le 99\}.$$

Also,

$$E[N] = \frac{500}{3}, \qquad \text{Var}(N) = 500 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{1000}{9}, \qquad \text{SD}(N) = \sqrt{\frac{1000}{9}}.$$

(a) Since $X \ge 0$, Markov's inequality applies:

$$P(X > 500) \le \frac{E[X]}{500} = \frac{300}{500} = \frac{3}{5}.$$

Final: $\boxed{P(X > 500) \le \frac{3}{5}}$

(b) Since $X > 500$ implies $X - 300 > 200$, we have

$$\{X > 500\} \subseteq \{|X - 300| \geq 200\}.$$

Hence Chebyshev's inequality gives

$$P(X > 500) \leq P(|X - 300| \geq 200) \leq \frac{\mathrm{Var}(X)}{200^2} = \frac{600}{40000} = \frac{3}{200} = 0.015.$$

Final: $\boxed{P(X > 500) \leq \frac{3}{200} = 0.015}$

(c) Using the representation $X = G_1 + \cdots + G_{100}$ and the CLT,

$$\frac{X - 300}{\sqrt{600}} \approx N(0, 1).$$

Therefore

$$P(X > 500) = P\left(\frac{X - 300}{\sqrt{600}} > \frac{500 - 300}{\sqrt{600}}\right) \approx 1 - \Phi\left(\frac{200}{\sqrt{600}}\right).$$

Since

$$\frac{200}{\sqrt{600}} = \frac{20}{\sqrt{6}} \approx 8.165,$$

we get

$$P(X > 500) \approx 1 - \Phi\left(\frac{20}{\sqrt{6}}\right) \approx 1.6 \times 10^{-16}.$$

Final: $\boxed{P(X > 500) \approx 1 - \Phi\left(\frac{20}{\sqrt{6}}\right) \approx 1.6 \times 10^{-16}}$

(d) Let N be the number of successes in the first 500 trials. Then

$$N \sim \mathrm{Bin}\left(500, \frac{1}{3}\right)$$

and

$$P(X > 500) = P(N \leq 99).$$

Using the CLT with a continuity correction,

$$P(N \leq 99) \approx P(N < 99.5) = P\left(\frac{N - 500/3}{\sqrt{1000/9}} < \frac{99.5 - 500/3}{\sqrt{1000/9}}\right) \approx \Phi\left(\frac{99.5 - 500/3}{\sqrt{1000/9}}\right).$$

Now

$$\frac{99.5 - 500/3}{\sqrt{1000/9}} \approx -6.372,$$

so

$$P(X > 500) = P(N \leq 99) \approx \Phi(-6.372) \approx 9.3 \times 10^{-11}.$$

This is different from part (c), which is not surprising: both are CLT approximations, and this is an extreme tail event.

Final: $\boxed{P(X > 500) = P(N \leq 99) \approx \Phi(-6.372) \approx 9.3 \times 10^{-11}}$